

Free-field Green's function

- Given a general solution to the homogeneous wave eq.: $p'(\mathbf{x}, t) = \int_{-T}^T \int_S \left(p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} - G(\mathbf{x}, t | \mathbf{y}, \tau) \frac{\partial p'(\mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau$
- We can now analyze **any radiating surface, given BCs**
- If there is an imposed flow v_i at the surface, then the pressure gradient term can be obtained from the acoustic momentum eq.:

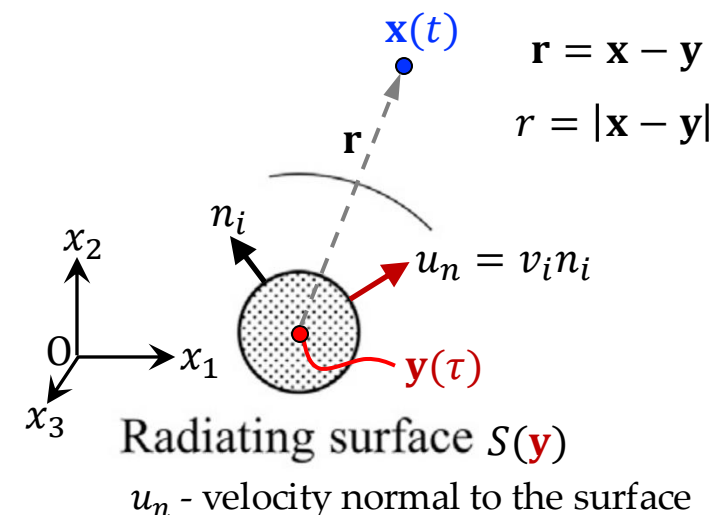
$$\text{Acoustic momentum eq.} \quad \rho_0 \frac{\partial v_i}{\partial t} + \frac{\partial p'}{\partial x_i} = 0 \quad \Rightarrow \quad \frac{\partial p'}{\partial y_i} n_i = \mathbf{n} \cdot \nabla p' = -\rho_0 \frac{\partial v_i}{\partial \tau} n_i$$

- If the fixed surface of integration is used to represent a **solid surface that is vibrating**, then the imposed flow can be taken to be the vibration velocity on the surface
- Therefore, we can obtain:

$$p'(\mathbf{x}, t) = \int_{-T}^T \int_S \left(p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau + \int_{-T}^T \int_S \left(\rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} G(\mathbf{x}, t | \mathbf{y}, \tau) \right) dS(\mathbf{y}) d\tau$$

Sound radiated from **force per unit area** $p'n_i$ exerted on fluid by surface

Sound radiated by **imposed velocity**

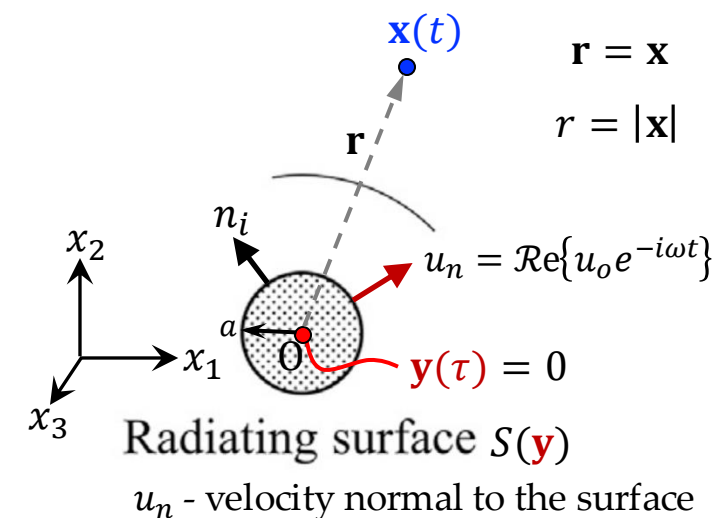


Free-field Green's function

- ❑ To evaluate the integrals, **we need to specify the Green's function**
- ❑ We already know the solution for a small **pulsating sphere**...
- ❑ Consider a small spherical surface with its center at $\mathbf{y} = 0$
 - Normal velocity on the surface is given by $u_n = \text{Re}\{u_o e^{-i\omega t}\}$
- ❑ **We allow the radius of the sphere source to be really small ($a \rightarrow 0$), so the spatial dependence of G and $\partial G / \partial \mathbf{y}_i$ is negligible**
 - In a sense, we assume a **compact sphere source** ($ka \ll 1$)
 - If observer is not on the sphere (i.e., $\mathbf{x} \neq \mathbf{y}$), and $G(\mathbf{x}|\mathbf{y})$ is a smooth function of $\mathbf{y}$, we can expand it via Taylor series about the center $\mathbf{y} = 0$:

$$G(\mathbf{x}|\mathbf{y}) = G(\mathbf{x}|0) + y_i \left. \frac{\partial G}{\partial y_i} \right|_{\mathbf{y}=0} + \mathcal{O}(|\mathbf{y}|^2) \quad \frac{\partial G(\mathbf{x}|\mathbf{y})}{\partial y_i} = \left. \frac{\partial G}{\partial y_i} \right|_{\mathbf{y}=0} + y_i \left. \frac{\partial^2 G}{\partial y_i \partial y_j} \right|_{\mathbf{y}=0} + \mathcal{O}(|\mathbf{y}|^2)$$

- On the sphere $|\mathbf{y}| = a$, and so we obtain: $\Delta G \sim a \left. \frac{\partial G}{\partial y_i} \right|_{\mathbf{y}=0}$ and $\Delta \left(\frac{\partial G}{\partial y_i} \right) \sim a \left. \frac{\partial^2 G}{\partial y_i \partial y_j} \right|_{\mathbf{y}=0}$
- But, as we take $a \rightarrow 0$, we obtain: $G(\mathbf{x}|\mathbf{y}) = \mathbb{C}$ and $\partial G / \partial y_i = \mathbb{C}$



Free-field Green's function

- Therefore, assuming a vanishingly small sphere, we can take G and $\partial G / \partial y_i$ out of the surface integral (assuming they are constants), yielding:

$$p'(\mathbf{x}, t) = \int_{-T}^T \int_S \left(p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau + \int_{-T}^T \int_S \left(\rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} G(\mathbf{x}, t | \mathbf{y}, \tau) \right) dS(\mathbf{y}) d\tau$$

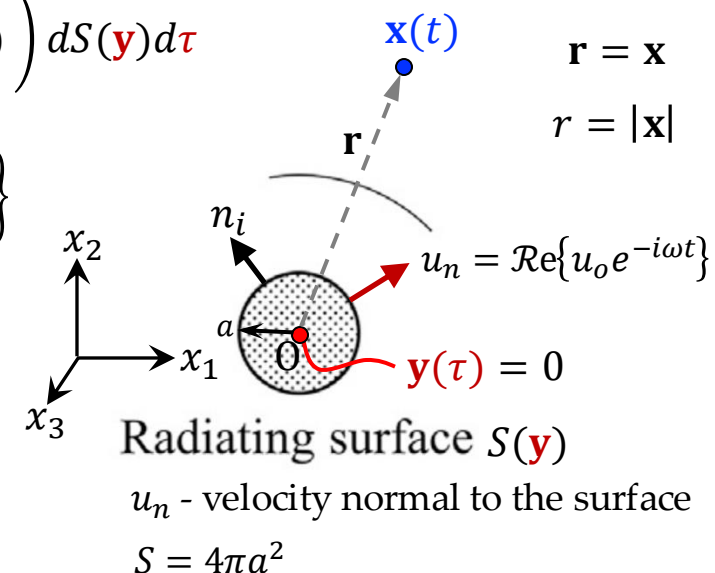
$$\Rightarrow p'(\mathbf{x}, t) = \int_{-T}^T \left[\frac{\partial G}{\partial y_i} \right]_{y_i=0} \int_S p'(\mathbf{y}, \tau) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau + \text{Re} \left\{ \int_{-T}^T [G]_{y_i=0} \int_S (-i\omega \rho_0 u_o e^{-i\omega\tau}) dS(\mathbf{y}) d\tau \right\}$$

Net force exerted on the fluid by the sphere is **zero** since pressure is **constant** on the surface

$$\int_S p' n_i dS = 0$$

$$\left[\frac{\partial p'}{\partial r} \right]_{r=a} = -\rho_0 \frac{\partial v_i}{\partial \tau} n_i = \text{Re} \{ i\omega \rho_0 u_o e^{-i\omega\tau} \} = \mathbb{C} \Rightarrow \int_S (-i\omega \rho_0 u_o e^{-i\omega\tau}) dS(\mathbf{y}) = -i\omega \rho_0 u_o S e^{-i\omega\tau}$$

$$p'(\mathbf{x}, t) = \text{Re} \left\{ \int_{-T}^T (-i\omega \rho_0 u_o S e^{-i\omega\tau}) [G(\mathbf{x}, t | \mathbf{y}, \tau)]_{y_i=0} d\tau \right\}$$



Free-field Green's function

- Let us **compare** the result we obtain to the **monopole sound field** we obtained previously

$$p'(\mathbf{x}, t) = \text{Re} \left\{ \int_{-T}^T (-i\omega\rho_0 u_0 S e^{-i\omega\tau}) [G(\mathbf{x}, t | \mathbf{y}, \tau)]_{\mathbf{y}_i=0} d\tau \right\}$$

$$p'(r) = -\frac{i\omega\rho_0 u_0 S e^{-i(\omega t + kr)}}{4\pi r}$$

- We reveal **Free-Field Green's function** G_0

➤ Given $r = |\mathbf{x} - \mathbf{y}| > 0$

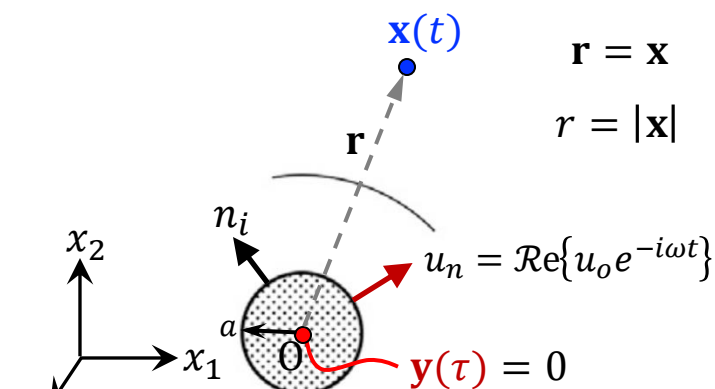
Source is located at $\mathbf{y} = 0$

$$G_0(r, t - \tau) = \frac{\delta(t - r/c_0 - \tau)}{4\pi r}$$

Source is located at $\mathbf{y}$

$$G_0(\mathbf{x}, t | \mathbf{y}, \tau) = \frac{\delta(t - |\mathbf{x} - \mathbf{y}|/c_0 - \tau)}{4\pi |\mathbf{x} - \mathbf{y}|}$$

- Acoustic field at the observer location $\mathbf{x}$ caused by an infinitely small volume displacement source (*monopole*) in an **infinite domain**



Radiating surface $S(\mathbf{y})$

u_n - velocity normal to the surface

$$S = 4\pi a^2$$

Monopole sound field

$$\hat{p}(r) = -\frac{i\omega\rho_0 u_0 S e^{ikr}}{4\pi r} \quad k = \omega/c_0$$

$$p'(r) = \hat{p}(r) e^{-i\omega t}$$

Free-field Green's function

- The **free-field Green's function** G_o is the acoustic field (solution of a wave eq.) generated at position $\mathbf{x}$ of an observer at time t by a pulse $\delta(\mathbf{x} - \mathbf{y})\delta(t - \tau)$ emitted at $\mathbf{y}$ at time τ , provided that $r = |\mathbf{x} - \mathbf{y}| > 0$

$$\mathbf{r} = \mathbf{x} - \mathbf{y}$$

$$r = |\mathbf{x} - \mathbf{y}|$$

$$G_o(\mathbf{x}, t | \mathbf{y}, \tau) = \frac{\delta(t - |\mathbf{x} - \mathbf{y}|/c_o - \tau)}{4\pi|\mathbf{x} - \mathbf{y}|}$$

is **A solution** to

$$\frac{1}{c_o^2} \frac{\partial^2 G_o}{\partial \tau^2} - \frac{\partial^2 G_o}{\partial y_i^2} = \delta(\mathbf{x} - \mathbf{y})\delta(t - \tau)$$

- The Green's free-field function G_o satisfies the **principle of causality** when $r > 0$

For $t < \tau$:

$$G_o = \frac{\partial G_o}{\partial \tau} = 0$$

because $r/c_o > 0$

- Since it depends only on r , it also satisfies the **symmetry property**:

$$\frac{\partial G_o}{\partial x_i} = - \frac{\partial G_o}{\partial y_i}$$

because $\frac{\partial r}{\partial x_i} = \frac{x_i - y_i}{r} = - \frac{\partial r}{\partial y_i}$

$$r = \sqrt{(x_i - y_i)^2}$$

Bringing the **observer** closer to the **source** has the same effect as bringing the **observer** closer to the **source**!

Free-field Green's function

$$G_o(\mathbf{x}, t | \mathbf{y}, \tau) = \frac{\delta(t - |\mathbf{x} - \mathbf{y}|/c_o - \tau)}{4\pi|\mathbf{x} - \mathbf{y}|}$$

□ Let us evaluate the spatial gradient of G_o

$$\frac{\partial G_o}{\partial y_i} = \frac{\partial}{\partial y_i} \left(\frac{\delta(t - |\mathbf{x} - \mathbf{y}|/c_o - \tau)}{4\pi|\mathbf{x} - \mathbf{y}|} \right) \Rightarrow \frac{\partial G_o}{\partial y_i} = \frac{1}{4\pi} \left[\underbrace{\frac{1}{r} \frac{\partial}{\partial y_i} (\delta(t - r/c_o - \tau))}_{\text{I}} + \underbrace{\delta(t - r/c_o - \tau) \frac{\partial}{\partial y_i} \left(\frac{1}{r} \right)}_{\text{II}} \right]$$

$$\text{I: } \frac{\partial}{\partial y_i} \left(\frac{1}{r} \right) = \frac{\partial r}{\partial y_i} \frac{\partial}{\partial r} \left(\frac{1}{r} \right) = -\frac{x_i - y_i}{r^2} \frac{\partial}{\partial r} \left(\frac{1}{r} \right) = \frac{x_i - y_i}{r^3}$$

$$\text{II: } \frac{\partial}{\partial y_i} (\delta(t - r/c_o - \tau)) = \frac{\partial r}{\partial y_i} \frac{\partial}{\partial r} (\delta(g(r))) = -\left(\frac{x_i - y_i}{r} \right) \frac{\partial}{\partial r} (\delta(g(r))) = -\left(\frac{x_i - y_i}{r} \right) \frac{\partial \delta}{\partial g} \frac{\partial g}{\partial r} = \left(\frac{x_i - y_i}{c_o r} \right) \dot{\delta}$$

$$\Rightarrow \frac{\partial G_o}{\partial y_i} = \frac{1}{4\pi} \left[\frac{1}{r} \left(\frac{x_i - y_i}{c_o r} \right) \dot{\delta} + \delta(t - r/c_o - \tau) \left(\frac{x_i - y_i}{r^3} \right) \right]$$

$$\frac{\partial G_o}{\partial y_i} = \left(\frac{x_i - y_i}{r} \right) \left(\frac{\dot{\delta} (t - r/c_o - \tau)}{4\pi r c_o} + \frac{\delta(t - r/c_o - \tau)}{4\pi r^2} \right)$$

- The field has a **cosine directionality** built in
- The field consists of both a **near-field** term and a **far-field** term

$$r = |\mathbf{x} - \mathbf{y}|$$

$$r = \sqrt{(x_i - y_i)^2}$$

$$\frac{\partial r}{\partial x_i} = \frac{x_i - y_i}{r} = -\frac{\partial r}{\partial y_i}$$

$$\dot{\delta} = \frac{\partial \delta}{\partial t} = \frac{\partial \delta}{\partial g}$$

$$g = t - r/c_o - \tau$$

Free-field Green's function

□ Let us use free-field Green's function to evaluate the time integral in the pressure equation $p'(\mathbf{x}, t)$

➤ We integrate the time derivative of the Dirac delta function *by parts*

$$\frac{\partial G_o}{\partial y_i} = \left(\frac{x_i - y_i}{r} \right) \left(\frac{\dot{\delta}(g)}{4\pi r c_o} + \frac{\delta(g)}{4\pi r^2} \right)$$

$$G_o(\mathbf{x}, t | \mathbf{y}, \tau) = \frac{\delta(g)}{4\pi r}$$

$$r = |\mathbf{x} - \mathbf{y}|$$

$$g = t - r/c_o - \tau$$

$$\dot{\delta} = \frac{\partial \delta}{\partial t}$$

$$p'(\mathbf{x}, t) = \underbrace{\int_{-T}^T \int_S \left(p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau}_{\text{I}} + \underbrace{\int_{-T}^T \int_S \left(\rho_o \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} G(\mathbf{x}, t | \mathbf{y}, \tau) \right) dS(\mathbf{y}) d\tau}_{\text{II}}$$

$$\text{II: } \int_{-T}^T \int_S \left(\rho_o \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} G(\mathbf{x}, t | \mathbf{y}, \tau) \right) dS(\mathbf{y}) d\tau \Rightarrow \int_S \int_{-T}^T \left(\frac{\rho_o}{4\pi r} \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \delta(g) \right) d\tau dS(\mathbf{y})$$

$$\Rightarrow \text{II} = \int_S \left[\rho_o \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right]_{\tau=\tau^*} \frac{dS(\mathbf{y})}{4\pi |\mathbf{x} - \mathbf{y}|}$$

Retarded time

$$\tau^* = t - \frac{|\mathbf{x} - \mathbf{y}|}{c_o}$$

Source/emission time

$$\int_{-\infty}^{\infty} f(\tau) \delta(t - r/c_o - \tau) d\tau = f(\tau^*)$$

Dirac delta function is **non-zero only at $\tau = \tau^*$**

Free-field Green's function

□ Let us use free-field Green's function to evaluate the time integral in the pressure equation $p'(\mathbf{x}, t)$

➤ We integrate the time derivative of the Dirac delta function *by parts*

$$\frac{\partial G_o}{\partial y_i} = \left(\frac{x_i - y_i}{r} \right) \left(\frac{\dot{\delta}(g)}{4\pi r c_o} + \frac{\delta(g)}{4\pi r^2} \right)$$

$$G_o(\mathbf{x}, t | \mathbf{y}, \tau) = \frac{\delta(g)}{4\pi r}$$

$$r = |\mathbf{x} - \mathbf{y}|$$

$$g = t - r/c_o - \tau$$

$$\dot{\delta} = \frac{\partial \delta}{\partial t}$$

$$p'(\mathbf{x}, t) = \underbrace{\int_{-T}^T \int_S \left(p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau}_{\text{I}} + \underbrace{\int_{-T}^T \int_S \left(\rho_o \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} G(\mathbf{x}, t | \mathbf{y}, \tau) \right) dS(\mathbf{y}) d\tau}_{\text{II}}$$

$$\text{I: } \int_{-T}^T \int_S \left(p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} \right) n_i dS(\mathbf{y}) d\tau \Rightarrow \int_S \int_{-T}^T p'(\mathbf{y}, \tau) n_i \left(\frac{x_i - y_i}{r} \right) \left(\frac{\dot{\delta}(g)}{4\pi r c_o} + \frac{\delta(g)}{4\pi r^2} \right) d\tau dS(\mathbf{y})$$

$$\Rightarrow \int_S n_i \left(\frac{x_i - y_i}{4\pi r^2} \right) \left[\frac{1}{c_o} \int_{-T}^T p'(\mathbf{y}, \tau) \dot{\delta}(g) d\tau + \frac{1}{r} \int_{-T}^T p'(\mathbf{y}, \tau) \delta(g) d\tau \right] dS(\mathbf{y})$$

Retarded time

$$\tau^* = t - \frac{|\mathbf{x} - \mathbf{y}|}{c_o}$$

Source/emission time

$$\int_{-\infty}^{\infty} f(\tau) \delta(t - r/c_o - \tau) d\tau = f(\tau^*)$$

Dirac delta function is **non-zero only at $\tau = \tau^*$**

Free-field Green's function

$$\mathbf{I}: \int_{-T}^T \int_S \left(p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} \right) n_i dS(\mathbf{y}) d\tau \Rightarrow \int_S n_i \left(\frac{x_i - y_i}{4\pi r^2} \right) \left[\frac{1}{c_o} \int_{-T}^T p'(\mathbf{y}, \tau) \dot{\delta}(g) d\tau + \frac{1}{r} \int_{-T}^T p'(\mathbf{y}, \tau) \delta(g) d\tau \right] dS(\mathbf{y})$$

$$r = |\mathbf{x} - \mathbf{y}|$$

$$g = t - r/c_o - \tau$$

$$\dot{\delta} = \frac{\partial \delta}{\partial t}$$

$$\xrightarrow{\text{Using sifting property}} \int_{-T}^T p'(\mathbf{y}, \tau) \delta(g) d\tau = [p'(\mathbf{y}, \tau)]_{\tau=\tau^*} \quad \text{We also use: } \dot{\delta} = \frac{\partial \delta}{\partial t} = -\frac{\partial \delta}{\partial \tau}$$

$$\Rightarrow \int_S n_i \left(\frac{x_i - y_i}{4\pi r^2} \right) \left[-\frac{1}{c_o} \int_{-T}^T p'(\mathbf{y}, \tau) \frac{\partial \delta(g)}{\partial \tau} d\tau + \frac{1}{r} [p'(\mathbf{y}, \tau)]_{\tau=\tau^*} \right] dS(\mathbf{y})$$

$$\text{Time-integration by parts: } \int u dv = uv - \int v du$$

$$\Rightarrow \int_S n_i \left(\frac{x_i - y_i}{4\pi r^2} \right) \left[-\frac{1}{c_o} \underbrace{[p'(\mathbf{y}, \tau) \delta(g)]_{-T}^T}_{\delta(g)=0 \text{ for } g \neq 0} + \frac{1}{c_o} \int_{-T}^T \delta(g) \frac{\partial p'(\mathbf{y}, \tau)}{\partial \tau} d\tau + \frac{1}{r} [p'(\mathbf{y}, \tau)]_{\tau=\tau^*} \right] dS(\mathbf{y})$$

Delta is zero outside a small neighborhood of $\tau = \tau^*$

$$\delta(g) = 0 \text{ for } g \neq 0$$

$$\Rightarrow \mathbf{I} = \int_S n_i \left(\frac{x_i - y_i}{4\pi r^2} \right) \left[\frac{1}{c_o} \frac{\partial p'(\mathbf{y}, \tau)}{\partial \tau} + \frac{1}{r} p'(\mathbf{y}, \tau) \right]_{\tau=\tau^*} dS(\mathbf{y})$$

Retarded time

$$\tau^* = t - \frac{|\mathbf{x} - \mathbf{y}|}{c_o}$$

Source/emission time

$$\int_{-\infty}^{\infty} f(\tau) \delta(t - r/c_o - \tau) d\tau = f(\tau^*)$$

Dirac delta function is **non-zero only at $\tau = \tau^*$**

Free-field Green's function

$$\tau^* = t - \frac{|\mathbf{x} - \mathbf{y}|}{c_o}$$

- Combining terms **I** and **II**, we can evaluate the time integral in the pressure equation $p'(\mathbf{x}, t)$

$$p'(\mathbf{x}, t) = \text{I} + \text{II} = \underbrace{\int_S \left[\frac{\partial p'(\mathbf{y}, \tau)}{\partial \tau} n_i + \frac{p'(\mathbf{y}, \tau) n_i c_o}{|\mathbf{x} - \mathbf{y}|} \right]_{\tau=\tau^*} \frac{(\mathbf{x}_i - \mathbf{y}_i) dS(\mathbf{y})}{4\pi |\mathbf{x} - \mathbf{y}|^2 c_o}}_{\text{Sound generated by surface loading}} + \underbrace{\int_S \left[\rho_o \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right]_{\tau=\tau^*} \frac{dS(\mathbf{y})}{4\pi |\mathbf{x} - \mathbf{y}|}}_{\text{Sound generated by imposed flow at the surface}}$$

PAY ATTENTION!

- The acoustic field exterior to the sources ($r = |\mathbf{x} - \mathbf{y}| > 0$) satisfies the **homogeneous wave eq.**
- The solution can be expressed using the **free-field Green's function** together with **appropriate BCs applied on the source surfaces** (Green's boundary integral formulation)

$$G_o(\mathbf{x}, t | \mathbf{y}, \tau) = \frac{\delta(t - |\mathbf{x} - \mathbf{y}|/c_o - \tau)}{4\pi |\mathbf{x} - \mathbf{y}|}$$

Free-field Green's function

□ In the acoustic far-field, $|\mathbf{x}| \gg |\mathbf{y}|$

$$p'(\mathbf{x}, t) = \int_S \left[\frac{\partial p'(\mathbf{y}, \tau)}{\partial \tau} n_i + \frac{p'(\mathbf{y}, \tau) n_i c_0}{|\mathbf{x} - \mathbf{y}|} \right]_{\tau=\tau^*} \frac{(\mathbf{x}_i - \mathbf{y}_i) dS(\mathbf{y})}{4\pi |\mathbf{x} - \mathbf{y}|^2 c_0} + \int_S \left[\rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right]_{\tau=\tau^*} \frac{dS(\mathbf{y})}{4\pi |\mathbf{x} - \mathbf{y}|}$$

$|\mathbf{x} - \mathbf{y}| \rightarrow |\mathbf{x}|$  $1/|\mathbf{x}|^3$ terms can be ignored

$$p'(\mathbf{x}, t) \approx \underbrace{\frac{\mathbf{x}_i}{4\pi |\mathbf{x}|^2 c_0} \int_S \left[\frac{\partial p'(\mathbf{y}, \tau)}{\partial \tau} n_i \right]_{\tau=\tau^*} dS(\mathbf{y})}_{\text{Dipole}} + \underbrace{\frac{1}{4\pi |\mathbf{x}|} \int_S \left[\rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right]_{\tau=\tau^*} dS(\mathbf{y})}_{\text{Monopole}}$$

Dipole

cosine directivity
(with peak in direction of net
force applied to fluid)

Monopole

Omnidirectional
(unless retarded time delays lead to
directionality; e.g., general surface motion)

The acoustic field is caused by
imposed velocity + forces exerted on the fluid at the surfaces

Solution of general spherical surface motions

- Notice that the solution we obtain for the pulsating sphere only accounts for **surface pressure** ($p'n_i$) as a source for the forces exerted on the fluid

$$p'(\mathbf{x}, t) = \int_{-T}^T \int_S \left(p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau + \int_{-T}^T \int_S \left(\rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} G(\mathbf{x}, t | \mathbf{y}, \tau) \right) dS(\mathbf{y}) d\tau$$

→ Sound radiated from force per unit area $p'n_i$ exerted on fluid by surface

→ Sound radiated by imposed velocity

- However, analyzing the case of a **translating sphere** shows that the **total force** $\hat{\mathbf{F}}$ applied to the fluid will be the **sum of**:

- Force required to **move the mass of fluid displaced by the sphere** ($m = \rho_0 V = \rho_0 \frac{4}{3} \pi a^3$)
- Force required to **overcome the net surface pressure** $\hat{p}$

$$\hat{\mathbf{F}} = -i\omega\rho_0 v_0 \left(\frac{4\pi a^3}{3} \right) \mathbf{n}_r + \int_S [\hat{p}]_{r=a} \mathbf{n}_r dS$$

How do we account for the *missing added mass effect*?

Solution of general spherical surface motions

□ We need to perform an asymptotic expansion of the Green's function in the limit of $\mathbf{y} \rightarrow 0$

$$G(\mathbf{x}, t | \mathbf{y}, \tau) = [G]_{\mathbf{y}=0} + y_i \left. \frac{\partial G}{\partial y_i} \right|_{\mathbf{y}=0} + \frac{y_i y_j}{2} \left. \frac{\partial^2 G}{\partial y_i \partial y_j} \right|_{\mathbf{y}=0} + \mathcal{O}(|\mathbf{y}|^3)$$

$$\frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} = \left. \frac{\partial G}{\partial y_i} \right|_{\mathbf{y}=0} + y_j \left. \frac{\partial^2 G}{\partial y_i \partial y_j} \right|_{\mathbf{y}=0} + \mathcal{O}(|\mathbf{y}|^2)$$

Substitute
to here

$$p'(\mathbf{x}, t) = \int_{-T}^T \int_S \left(p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau + \int_{-T}^T \int_S \left(\rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} G(\mathbf{x}, t | \mathbf{y}, \tau) \right) dS(\mathbf{y}) d\tau$$

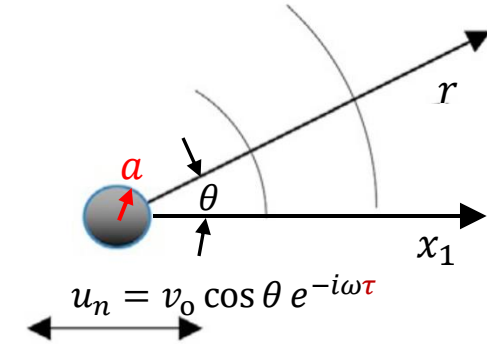
$$p'(\mathbf{x}, t) = \int_{-T}^T G(\mathbf{x}, t | 0, \tau) \int_S \left(\rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right) dS(\mathbf{y}) d\tau + \int_{-T}^T \frac{\partial G(\mathbf{x}, t | 0, \tau)}{\partial y_i} \int_S \left(p'(\mathbf{y}, \tau) n_i + y_i \rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right) dS(\mathbf{y}) d\tau + \int_{-T}^T \frac{\partial^2 G(\mathbf{x}, t | 0, \tau)}{\partial y_i \partial y_j} \int_S \left(p'(\mathbf{y}, \tau) n_i + \frac{y_i \rho_0}{2} \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right) y_j dS(\mathbf{y}) d\tau$$

Monopole
Net volumetric velocity, Q

Dipole
Net pressure, F

Quadrupole
No net volumetric velocity or pressure

Revisiting the translating sphere case



- Consider a translating sphere of radius a whose surface velocity is given by $u_n = v_0 \cos \theta e^{-i\omega\tau}$ on the spherical surface $|\mathbf{y}| = a$

- The solution to general spherical surface motions presented in the previous slide shows:

$$p'(\mathbf{x}, t) = \int_{-T}^T G(\mathbf{x}, t|0, \tau) \int_S \left(\rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right) dS(\mathbf{y}) d\tau + \int_{-T}^T \frac{\partial G(\mathbf{x}, t|0, \tau)}{\partial y_i} \int_S \left(p'(\mathbf{y}, \tau) n_i + y_i \rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right) dS(\mathbf{y}) d\tau + \int_{-T}^T \frac{\partial^2 G(\mathbf{x}, t|0, \tau)}{\partial y_i \partial y_j} \int_S \left(p'(\mathbf{y}, \tau) n_i + \frac{y_i \rho_0}{2} \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right) y_j dS(\mathbf{y}) d\tau + \dots$$

Dipole has no net volumetric velocity

Focus on the 2nd surface integral

Integral over $\cos \theta n_i n_j$ cancels out
 $\mathbf{y} = (r \cos \theta, r \sin \theta \cos \varphi, r \sin \theta \sin \varphi)$

$$\left(p'(\mathbf{y}, \tau) n_i + y_i \rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right) \xrightarrow[\text{on the sphere}]{y_i = a n_i} \boxed{(p'(\mathbf{y}, \tau) - i\omega a \rho_0 v_0 \cos \theta e^{-i\omega\tau}) n_i}$$

- Recall the result we obtained for the BC above with $ka \ll 1$:

$$\hat{p}(r) \equiv \frac{ika \cos \theta}{2} \left(\frac{i\omega \rho_0 v_0 a^2 e^{ikr}}{r} \right) \left(1 - \frac{1}{ikr} \right)$$

- Substituting the acoustic field we previously obtained for $r = a$:

$$(p'(\mathbf{y}, \tau) - i\omega a \rho_0 v_0 \cos \theta e^{-i\omega\tau}) n_i \Rightarrow - \left(\left[\frac{i\omega \rho_0 v_0 a \cos \theta}{2} \right]_{\text{attached mass}} + [i\omega a \rho_0 v_0 \cos \theta]_{\text{added mass}} \right) n_i e^{-i\omega\tau} = - \left(\frac{3}{2} i\omega a \rho_0 v_0 \cos \theta \right) n_i e^{-i\omega\tau}$$

$$\text{Recall previous result: } \hat{\mathbf{F}} = \int_S [\hat{p}]_{r=a} \mathbf{n}_r dS - i\omega \rho_0 v_0 \left(\frac{4\pi a^3}{3} \right) = -\frac{4\pi}{3} i\omega \rho_0 v_0 a^3 \left[\frac{1}{2} + 1 \right]$$

Higher-order expansion of G must be done to account for the added mass term!

Revisiting the translating sphere case

□ Substituting and using the free-field Green's function yields:

$$p'(\mathbf{x}, t) = \int_{-T}^T \frac{\partial G_o(\mathbf{x}, t | 0, \tau)}{\partial y_i} \int_S \left(p'(\mathbf{y}, \tau) n_i + y_i \rho_0 \frac{\partial u_n(\mathbf{y}, \tau)}{\partial \tau} \right) dS(\mathbf{y}) d\tau$$

$$\Rightarrow p'(\mathbf{x}, t) = - \int_{-T}^T \left(\frac{x_i}{|\mathbf{x}|} \right) \left(\frac{\dot{\delta}(g)}{4\pi|\mathbf{x}|c_o} + \frac{\delta(g)}{4\pi|\mathbf{x}|^2} \right) \int_S \left(\frac{3}{2} i\omega a \rho_0 v_o \cos \theta \right) n_i e^{-i\omega\tau} dS(\mathbf{y}) d\tau$$

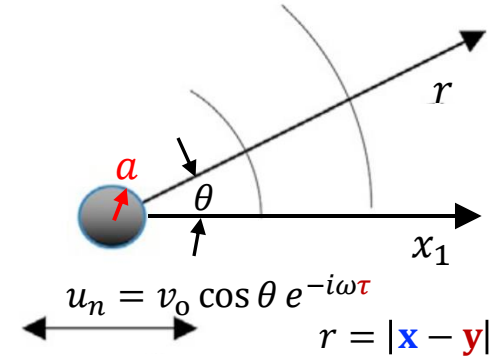
$$\Rightarrow p'(\mathbf{x}, t) = -\frac{3}{2} i\omega a \rho_0 v_o \left(\frac{x_i}{4\pi|\mathbf{x}|^2} \right) \int_{-T}^T \left(\frac{\dot{\delta}(g)}{c_o} + \frac{\delta(g)}{|\mathbf{x}|} \right) e^{-i\omega\tau} \int_0^\pi \int_0^{2\pi} \cos \theta \cos \theta a^2 \sin \theta d\theta d\varphi d\tau$$

$$\Rightarrow p'(\mathbf{x}, t) = -3\pi i\omega a^3 \rho_0 v_o \left(\frac{x_i}{4\pi|\mathbf{x}|^2} \right) \int_{-T}^T \left(\frac{\dot{\delta}(g)}{c_o} + \frac{\delta(g)}{|\mathbf{x}|} \right) e^{-i\omega\tau} \underbrace{\int_0^\pi \cos^2 \theta \sin \theta d\theta}_{= 2/3} d\tau$$

$$\Rightarrow p'(\mathbf{x}, t) = -2\pi i\omega a^3 \rho_0 v_o \left(\frac{x_i}{4\pi|\mathbf{x}|^2} \right) \left[-\frac{1}{c_o} \int_{-T}^T e^{-i\omega\tau} \frac{\partial \delta(g)}{\partial \tau} d\tau + \frac{1}{|\mathbf{x}|} \int_{-T}^T e^{-i\omega\tau} \delta(g) d\tau \right]$$

$$\Rightarrow p'(\mathbf{x}, t) = -2\pi i\omega a^3 \rho_0 v_o \left(\frac{x_i}{4\pi|\mathbf{x}|^2} \right) \left[-\frac{1}{c_o} \underbrace{\left[e^{-i\omega\tau} \delta(g) \right]_{-T}^T}_{=0} - \frac{i\omega}{c_o} \int_{-T}^T \delta(g) e^{-i\omega\tau} d\tau + \frac{1}{|\mathbf{x}|} e^{-i\omega\tau^*} \right]$$

Delta is zero outside a small neighborhood of $\tau = \tau^*$ $\delta(g) = 0$ for $g \neq 0$



$$\mathbf{y} = (r \cos \theta, r \sin \theta \cos \varphi, r \sin \theta \sin \varphi)$$

$$g = t - |\mathbf{x} - \mathbf{y}|/c_o - \tau$$

$$G_o = \frac{\delta(g)}{4\pi|\mathbf{x} - \mathbf{y}|}$$

$$\frac{\partial G_o}{\partial y_i} = \left(\frac{x_i - y_i}{r} \right) \left(\frac{\dot{\delta}(g)}{4\pi r c_o} + \frac{\delta(g)}{4\pi r^2} \right)$$

$$\dot{\delta} = \frac{\partial \delta}{\partial t} = -\frac{\partial \delta}{\partial \tau}$$

Time-integration by parts: $\int u dv = uv - \int v du$

$$\int_{-\infty}^{\infty} f(\tau) \delta(t - r/c_o - \tau) d\tau = f(\tau^*)$$

Dirac delta function is non-zero only at $\tau = \tau^*$

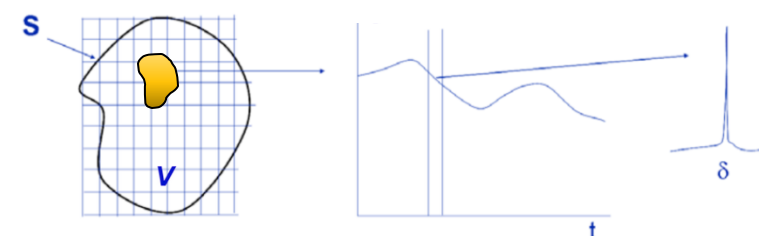
$$\tau^* = t - |\mathbf{x} - \mathbf{y}|/c_o$$

Alternative solution in the presence of surfaces via inhomogeneous wave eq.

- Alternative approach to solve noise radiation from surfaces involves solving the **inhomogeneous wave eq.** with the BCs imposed via a **source term** $\mathcal{S}(\mathbf{y}, \tau)$ and an **initial conditions*** ($\tau = t_0$)

$$\frac{1}{c_0^2} \frac{\partial^2 p'(\mathbf{y}, \tau)}{\partial \tau^2} - \frac{\partial^2 p'(\mathbf{y}, \tau)}{\partial y_i^2} = \frac{\partial Q(\mathbf{y}, \tau)}{\partial \tau} - \frac{\partial F_i(\mathbf{y}, \tau)}{\partial y_i} = \mathcal{S}(\mathbf{y}, \tau)$$

Using a similar approach shown before



$$p'(\mathbf{x}, t) = \int_{t_0}^t \int_V \mathcal{S}(\mathbf{y}, \tau) G(\mathbf{x}, t | \mathbf{y}, \tau) dV d\tau$$

Convolution of the source term $\mathcal{S}$ and G

$$+ \int_{t_0}^t \int_S \left(p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} \right.$$

Difference between BCs on the surface and G

$$\left. - G(\mathbf{x}, t | \mathbf{y}, \tau) \frac{\partial p'(\mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau$$

Initial conditions at t_0

$$- \frac{1}{c_0^2} \int_V \left[p'(\mathbf{y}, t_0) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial \tau} - G(\mathbf{x}, t | \mathbf{y}, \tau) \frac{\partial p'(\mathbf{y}, \tau)}{\partial \tau} \right]_{\tau=t_0} dV$$

$$p'(\mathbf{y}, t_0)$$

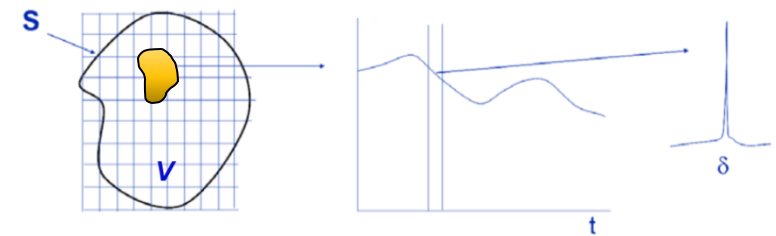
$$\partial p'(\mathbf{y}, t_0) / \partial \tau$$

*causality is still enforced

Alternative solution in the presence of surfaces via *inhomogeneous* wave eq.

□ Note that for **canonical problems** (point monopole, dipole,...) we usually assume:

- Fluid initially at rest: $p'(\mathbf{y}, t_0) = 0$, $\partial p'(\mathbf{y}, t_0)/\partial \tau = 0$
- Source surface is modeled as a **point source at the origin** ($\mathbf{y} = 0$)
 - Acoustically compact...



□ Therefore, we only left with the **convolution term**:

$$p'(\mathbf{x}, t) = \int_{t_0}^t \int_V \mathcal{S}(\mathbf{y}, \tau) G(\mathbf{x}, t | \mathbf{y}, \tau) dV d\tau$$

$$+ \int_{t_0}^t \int_S \left(p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial y_i} - G(\mathbf{x}, t | \mathbf{y}, \tau) \frac{\partial p'(\mathbf{y}, \tau)}{\partial y_i} \right) n_i(\mathbf{y}) dS(\mathbf{y}) d\tau$$

$$- \frac{1}{c_0^2} \int_V \left[p'(\mathbf{y}, \tau) \frac{\partial G(\mathbf{x}, t | \mathbf{y}, \tau)}{\partial \tau} - G(\mathbf{x}, t | \mathbf{y}, \tau) \frac{\partial p'(\mathbf{y}, \tau)}{\partial \tau} \right]_{\tau=t} dV$$

Alternative solution in the presence of surfaces via inhomogeneous wave eq.

□ Example – pulsating sphere at ($\mathbf{y} = 0$)

- Net volumetric velocity exist (Q), but no net force ($F_i = 0$)

$$\mathcal{S}(\mathbf{y}, \tau) = \frac{\partial Q(\mathbf{y}, \tau)}{\partial \tau} \quad \boxed{Q(\tau) = u_o S e^{-i\omega\tau}} \quad \hat{Q} = u_o S$$

- Treat the sphere as acoustically compact ($ka \ll 1$), and model it as a **point monopole at the origin** ($\mathbf{y} = 0$)

$$Q(\mathbf{y}, \tau) = \rho_o Q(\tau) \delta(\mathbf{y}) \Rightarrow \mathcal{S}(\mathbf{y}, \tau) = \rho_o \frac{\partial Q(\tau)}{\partial \tau} \delta(\mathbf{y})$$

- Therefore, we obtain:

$$p'(\mathbf{x}, t) = \rho_o \int_{t_0}^t \int_V \frac{\partial Q(\tau)}{\partial \tau} \delta(\mathbf{y}) G(\mathbf{x}, t | \mathbf{y}, \tau) dV d\tau \Rightarrow p'(\mathbf{x}, t) = \rho_o \int_{t_0}^t \frac{\partial Q(\tau)}{\partial \tau} \frac{\delta(g)}{4\pi|\mathbf{x}|} d\tau \Rightarrow p'(\mathbf{x}, t) = \frac{\rho_o}{4\pi|\mathbf{x}|} \int_{t_0}^t \frac{\partial Q(\tau)}{\partial \tau} \delta(g) d\tau$$

$$\Rightarrow p'(\mathbf{x}, t) = \frac{\rho_o}{4\pi|\mathbf{x}|} \left[\frac{\partial Q(\tau)}{\partial \tau} \right]_{\tau=\tau^*} \Rightarrow p'(\mathbf{x}, t) = -i\omega \rho_o \hat{Q} \frac{e^{-i\omega\tau^*}}{4\pi|\mathbf{x}|}$$

$$\Rightarrow p'(\mathbf{x}, t) = -i\omega \rho_o \hat{Q} \frac{e^{-i(\omega t - kr)}}{4\pi|\mathbf{x}|}$$

Identical to what we found previously

$$\hat{p}(r) = -i\omega \rho_o \hat{Q} \frac{e^{ikr}}{4\pi r}$$

$$p'(\mathbf{x}, t) = \int_{t_0}^t \int_V \mathcal{S}(\mathbf{y}, \tau) G(\mathbf{x}, t | \mathbf{y}, \tau) dV d\tau$$

$$\tau^* = t - |\mathbf{x}|/c_o \quad g = t - |\mathbf{x}|/c_o - \tau$$

$$r = |\mathbf{x}| \quad G_o(\mathbf{x}, t | 0, \tau) = \frac{\delta(g)}{4\pi|\mathbf{x}|}$$

Alternative solution in the presence of surfaces via inhomogeneous wave eq.

□ Example – translating sphere at ($\mathbf{y} = 0$)

- No net volumetric velocity exist ($Q = 0$), but a net force exist (F_i)

$$\mathcal{S}(\mathbf{y}, \tau) = -\frac{\partial F_i(\mathbf{y}, \tau)}{\partial y_i} \quad \boxed{F_i(\tau) = \hat{F}_i e^{-i\omega\tau}}$$

- Treat the sphere as acoustically compact ($ka \ll 1$), and model it as a **point dipole at the origin** ($\mathbf{y} = 0$)

$$F_i(\mathbf{y}, \tau) = F_i(\tau)\delta(\mathbf{y}) \Rightarrow \mathcal{S}(\mathbf{y}, \tau) = -F_i(\tau) \frac{\partial \delta(\mathbf{y})}{\partial y_i}$$

- We obtain: $p'(\mathbf{x}, t) = -\int_{t_0}^t \int_V F_i(\tau) \frac{\partial \delta(\mathbf{y})}{\partial y_i} G(\mathbf{x}, t|\mathbf{y}, \tau) dV d\tau \xrightarrow[\text{by parts in } \mathbf{y}]{\text{integration}} p'(\mathbf{x}, t) = \int_{t_0}^t F_i(\tau) \frac{\partial G(\mathbf{x}, t|0, \tau)}{\partial y_i} d\tau$

$$\Rightarrow p'(\mathbf{x}, t) = \frac{x_i}{4\pi|\mathbf{x}|^2} \int_{t_0}^t \left(-\frac{F_i(\tau)}{c_0} \frac{\partial \delta(g)}{\partial \tau} + \frac{F_i(\tau)\delta(g)}{|\mathbf{x}|} \right) d\tau \xrightarrow[\text{by part for } \delta]{\text{integration}} p'(\mathbf{x}, t) = \frac{x_i}{4\pi|\mathbf{x}|^2} \left[-\frac{1}{c_0} \underbrace{[F_i(\tau)\delta(g)]_{-T}^T}_{\delta(g)=0 \text{ for } g \neq 0} + \int_{t_0}^t \frac{\delta(g)}{c_0} \frac{\partial F_i(\tau)}{\partial \tau} d\tau + \frac{1}{|\mathbf{x}|} F_i(\tau^*) \right]$$

$$\Rightarrow p'(\mathbf{x}, t) = \frac{x_i}{4\pi|\mathbf{x}|^2} \left[\frac{1}{c_0} \frac{\partial F_i(\tau^*)}{\partial \tau} + \frac{1}{|\mathbf{x}|} F_i(\tau^*) \right] \Rightarrow p'(\mathbf{x}, t) = \frac{x_i}{4\pi|\mathbf{x}|^2} \hat{F}_i \left[-ik + \frac{1}{|\mathbf{x}|} \right] e^{-i\omega\tau^*}$$

$$p'(\mathbf{x}, t) = -ik \frac{x_i}{|\mathbf{x}|} \left(\frac{\hat{F}_i}{4\pi|\mathbf{x}|} \right) \left[1 - \frac{1}{ik|\mathbf{x}|} \right] e^{-i(\omega t - kr)}$$

Identical to what we found previously

$$\hat{p}(r) = -ik \cos \theta e^{ikr} \left(\frac{\hat{F}_1}{4\pi r} \right) \left(1 - \frac{1}{ikr} \right)$$

$$p'(\mathbf{x}, t) = \int_{t_0}^t \int_V \mathcal{S}(\mathbf{y}, \tau) G(\mathbf{x}, t|\mathbf{y}, \tau) dV d\tau$$

$$\tau^* = t - |\mathbf{x}|/c_0 \quad g = t - |\mathbf{x}|/c_0 - \tau$$

$$r = |\mathbf{x}| \quad G_0(\mathbf{x}, t|0, \tau) = \frac{\delta(g)}{4\pi|\mathbf{x}|}$$

$$\frac{\partial G_0}{\partial y_i} = \left(\frac{x_i}{|\mathbf{x}|} \right) \left(\frac{\dot{\delta}(g)}{4\pi|\mathbf{x}|c_0} + \frac{\delta(g)}{4\pi|\mathbf{x}|^2} \right)$$

$$\dot{\delta} = \frac{\partial \delta}{\partial t} = -\frac{\partial \delta}{\partial \tau}$$

Summary

□ Noise radiation from elementary sources can be analyzed in two *equivalent* methods

➤ **Free-field Green's function**

- **Green's boundary integral formulation** – appropriate BCs applied on the source surfaces, and the *homogeneous wave equation* is solved in the fluid (exterior) region
- **Equivalent sources view** – sources are introduced *locally* as either a (monopole) volumetric flow rate $Q(\mathbf{x}, t)$ or a (dipole) body force $F_i(\mathbf{x}, t)$, leading to an *inhomogeneous wave equation* with a source term $\mathcal{S}$

$$\frac{1}{c_0^2} \frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 p'}{\partial x_i^2} = \frac{\partial Q}{\partial t} - \frac{\partial F_i}{\partial x_i} = \mathcal{S}(\mathbf{x}, t)$$

or

$$\frac{\partial^2 \hat{p}}{\partial x_i^2} + k^2 \hat{p} = -i\omega \hat{Q} - \frac{\partial \hat{F}}{\partial x_i} = \hat{\mathcal{S}}(\mathbf{x}, \omega)$$

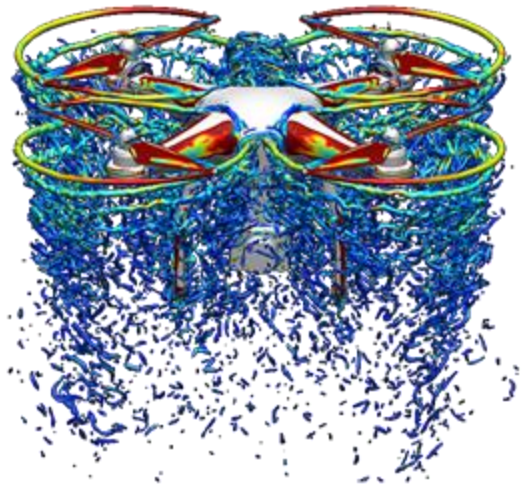
➤ **Direct solution of the 2nd-order *linear* homogeneous wave equation**

- Solution is achieved by imposing BCs on the actual surfaces, which determine the unknown **constants** introduced by the sources
- Multiple sources can be treated by **superposition** of their individual homogeneous solutions

$$\frac{1}{c_0^2} \frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 p'}{\partial x_i^2} = 0$$

or

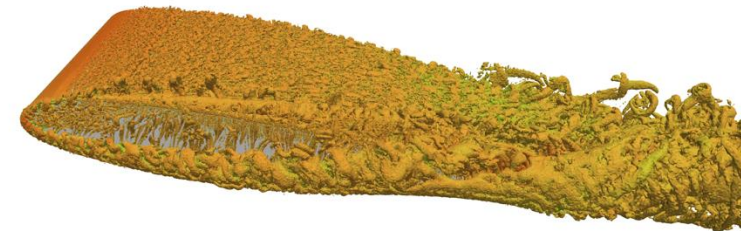
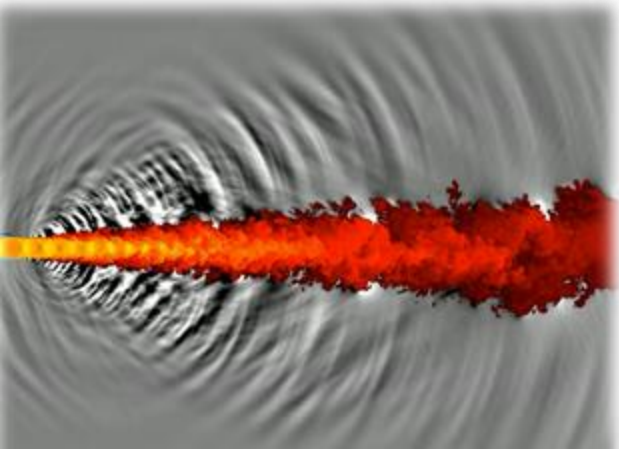
$$\frac{\partial^2 \hat{p}}{\partial x_i^2} + k^2 \hat{p} = 0$$



Chapter 3

Acoustic Analogies

$$\frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}$$

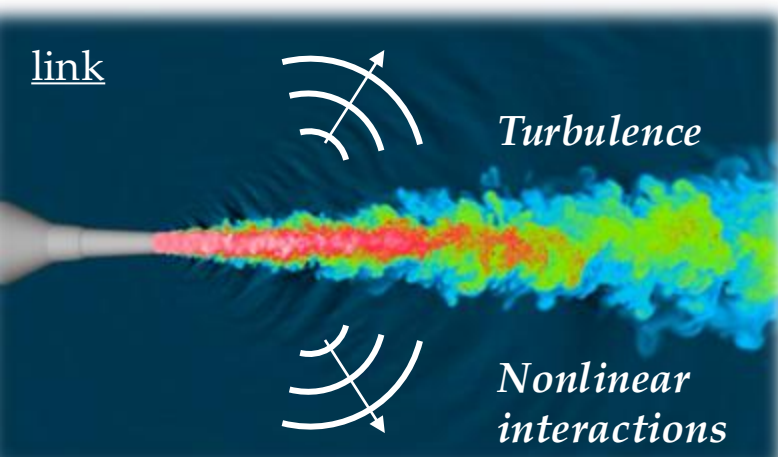


Shortcomings of the linear acoustics

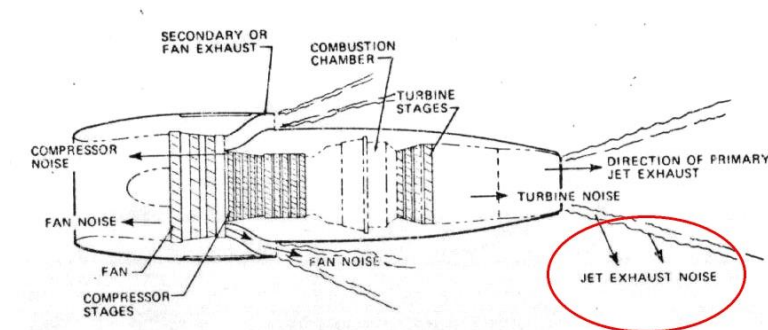
- ❑ We showed that, in **ideal conditions**, the propagation of sound in a **uniform stationary medium** can be described by the **linear acoustic wave equation** with p' as the dependent variable
 - Sound sources were generated by *disturbances* caused by a prescribed *body surface motion*
 - BCs were *matched* using the *linearized* momentum equation
- ❑ However, soon after the invention of the jet engine, it was realized that **linear acoustics cannot be used to predict jet noise**
- ❑ When a jet exhausts into a stationary fluid, a **large region of turbulent flow (high-Reynolds)** is generated, but there are **no surfaces with prescribed motions to produce sounds!**

Linear acoustics

$$p' \ll p_0 \quad \mathbf{v}' \ll c_0 \quad \rho' \ll \rho_0$$



Some other mechanisms for creating sound waves are at play...

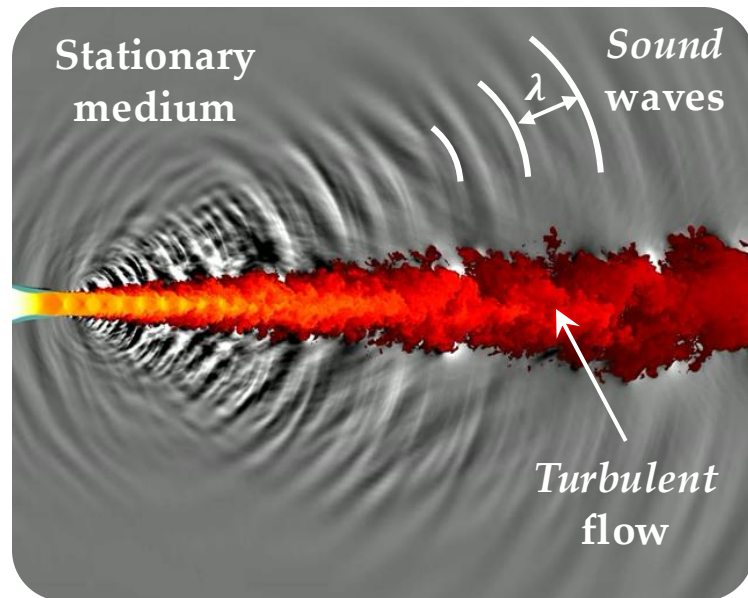


Lighthill's analogy

- ❑ **Lighthill's analogy** (1952) addresses sound generation by a region of **turbulent flow** in a **stationary fluid**
 - Motivated by the need to evaluate and mitigate jet noise
 - Forms the *basis of our understanding of aeroacoustics* (flow-induced sound)



Sir James Lighthill
(23/01/1924 – 17/07/1998)



Lighthill's objective
Separate terms describing the *generation* of sound waves *propagating* into the acoustic far field from the equations describing fluid motion in a turbulent flow

On sound generated aerodynamically
I. General theory
By M. J. LIGHTHILL
Department of Mathematics, The University of Manchester
(Communicated by M. H. A. Newman, F.R.S.—Received 13 November 1951)

A theory is initiated, based on the equations of motion of a gas, for the purpose of estimating the sound radiated from a fluid flow, with rigid boundaries, which as a result of instability contains regular fluctuations or turbulence. The sound field is that which would be produced by a static distribution of acoustic quadrupoles whose instantaneous strength per unit volume is $\rho \mathbf{v} \cdot \mathbf{v} - \frac{1}{2} \rho \mathbf{v}^2$, where ρ is the density, $\mathbf{v}$ the velocity vector, $\rho \mathbf{v} \cdot \mathbf{v}$ the compressive stress tensor, and $\mathbf{v} \cdot \mathbf{v}$ the velocity of sound outside the flow. This quadrupole strength density may be approximated in many cases as $\rho \mathbf{v} \cdot \mathbf{v}$. The radiation field is deduced by means of retarded potential solutions. In it, the intensity depends crucially on the frequency as well as on the strength of the quadrupoles, and as a result increases in proportion to a high power, near the eighth, of a typical velocity U in the flow. Physically, the mechanism of conversion of energy from kinetic to acoustic is based on fluctuations in the flow of momentum across fixed surfaces, and it is explained in §2 how this accounts both for the relative inefficiency of the process and for the increase of efficiency with U . It is shown in §7 how the efficiency is also increased, particularly for the sound emitted forwards, in the case of fluctuations convected at a not negligible Mach number.

On sound generated aerodynamically
II. Turbulence as a source of sound
By M. J. LIGHTHILL, F.R.S.
Department of Mathematics, The University of Manchester
(Received 21 July 1953)

The theory of sound generated aerodynamically is extended by taking into account the statistical properties of turbulent surfaces, from which the sound radiated (without the help of solid boundaries) is called aerodynamic noise. The theory is developed with special reference to the noise of jets, for which a detailed comparison with experiment is made (§5 for subsonic jets, §8 for supersonic ones).
The quadrupole distribution of part I (Lighthill 1952) is shown to behave (see §3) as if it were concentrated into independent point quadrupoles, one in each 'average eddy volume'. The sound field of each of these is determined, in favour of downstream emission, by the general downstream motion of the eddy, in accordance with the quadrupole convection theory of part I. This explains, for jet noise, the marked preference for downstream emission, and its increase with jet velocity. For jet velocities considerably greater than the atmospheric speed of sound, the 'Mach number of convection' M_c may exceed 1 in parts of the jet, and then the directional maximum for radiation from those parts of the jet is at an angle of $\sin^{-1}(M_c)$ to the axis (§5).
Although turbulence without any mean flow has an acoustic power output, which was calculated to a rough approximation from the expression of part I by Proudman (1951) (see also §4 below), nevertheless, turbulence of given intensity can generate more sound in the presence of a large mean flow (§5). This sound has a directional maximum at 45° (or slightly less, due to the quadrupole convection effect) to the shear layer. These results follow from the fact that the most important term in the rate of change of momentum flux is the product of the pressure and the rate of strain (see figure 2). The higher frequency sound from the heavily sheared mixing region close to the surface of a jet is found to be of this character. But the lower frequency sound from the fully turbulent core of the jet, further downstream, can be estimated satisfactorily (§7) from Proudman's results, which are here reinterpreted (§8) in terms of sound generated from combined fluctuations of pressure and rate of strain in the turbulence. The acoustic efficiency of the jet is of the order of magnitude $10^{-8} M^8$, where M is the jet Mach number.

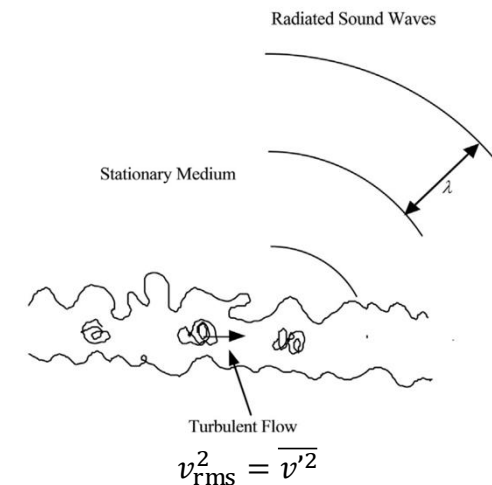
ly, that is, as a by-
vibration of solids,
ving at low Reynolds

Lighthill, M. J. (1952) "On sound generated aerodynamically I. General theory", *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 211(1107), pp. 564–587.

Lighthill, M. J. (1954) "On sound generated aerodynamically II. Turbulence as a source of sound", *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 222(1148), pp. 1–32.

Lighthill's analogy

Lighthill's challenge
Coupling between turbulent flow & acoustic waves ...



- ❑ Consider the following subsonic turbulent flow:
 - Mean velocity $v_o = 100$ m/s
 - Turbulence intensity, $TI = v_{rms}/v_o = 10\%$ (common)
- $v' \sim 10$ m/s
- ❑ Recall the result for **high-intensity** acoustic wave ($c_o = 343$ m/s) of SPL = 140 dB(re 20 μ Pa):

$$\frac{p_{rms}}{p_{ref}} = 10^{SPL/20} \Rightarrow \frac{p_{rms}}{p_{ref}} = 10^7 \Rightarrow \frac{p_{rms}}{p_o} = 10^7 \frac{p_{ref}}{p_o}$$

$$\begin{aligned} p_{rms} &\sim p' \\ p_{ref} &= 2 \cdot 10^{-5} \text{ Pa} \\ p_o &\approx 10^5 \text{ Pa} \\ p' &= c_o^2 \rho' \\ c_o^2 &= \gamma p_o / \rho_o \end{aligned}$$

$$\frac{p'}{p_o} = 10^7 \frac{2 \cdot 10^{-5}}{10^5} = 2 \cdot 10^{-3} \ll 1$$

$$\frac{\rho'}{\rho_o} = \frac{2 \cdot 10^{-3}}{1.4} \sim 10^{-3} \ll 1$$

- Acoustic wave particle velocity:

$$u' = \frac{p'}{\rho_o c_o} \xrightarrow[\text{flow, } p' = c_o^2 \rho']{\text{Homotropic}} u' = \frac{\rho' c_o}{\rho_o} \Rightarrow u' \approx 0.3 \text{ m/s}$$

**Acoustic wave is ~3% of the
unsteady motion (u'/v')**

CAUTION!

Lighthill's analogy

- ❑ Lighthill recognized he must be **cautious** about *coupling* the turbulent flow to the acoustic waves
 - Assuming *incompressible turbulence flow* results in **no sound!**
 - Assuming a *weakly compressible flow* permits **acoustic motion** in a fluid dominated by *incompressible turbulence*, but does this yield the correct result?

thus

Lighthill's Analogy

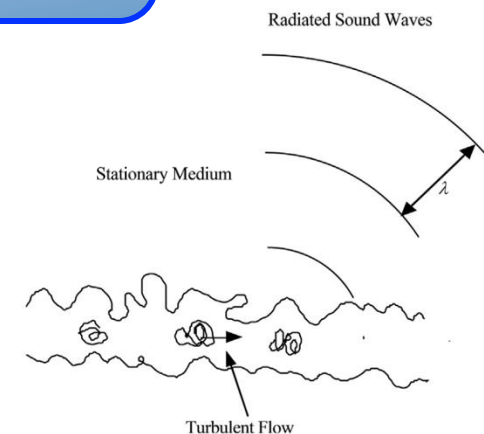
Based on the exact equations of fluid flow (full N-S eq.)

NO assumptions are made about compressibility effects in the turbulent region

NO linearization of the N-S equations!

Why is it so valuable?

Provides the **correct equations** for *coupling* the **acoustic wave motion** outside the turbulent region to the large velocity fluctuations inside the **turbulent flow**



Lighthill's analogy

- We start from the **compressible N-S equations**, taking the *time derivative* of the continuity eq. and subtracting the *divergence* of the momentum eq.

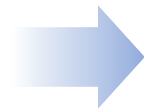
- Mass conservation

$$\frac{\partial}{\partial t} \left(\frac{\partial \rho}{\partial t} + \frac{\partial(\rho v_i)}{\partial x_i} \right) = 0$$

- Momentum conservation

$$\frac{\partial}{\partial x_i} \left(\frac{\partial(\rho v_i)}{\partial t} + \frac{\partial(\rho v_i v_j + p_{ij})}{\partial x_j} \right) = 0$$

(-) Subtract



$$\frac{\partial^2 \rho}{\partial t^2} = \frac{\partial^2(\rho v_i v_j + p_{ij})}{\partial x_i \partial x_j}$$

- Compressive stress tensor p_{ij} is defined relative to a *gauge pressure*: $p_{ij} = (p - p_\infty)\delta_{ij} - \sigma_{ij}$
 - p_∞ – surrounding ambient pressure
 - σ_{ij} – viscous shear stress tensor $\sigma_{ij} = \mu \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3} \frac{\partial v_k}{\partial x_k} \delta_{ij} \right)$

Lighthill's analogy

$$\frac{\partial^2 \rho}{\partial t^2} = \frac{\partial^2 (\rho v_i v_j + p_{ij})}{\partial x_i \partial x_j}$$

- We define the density perturbation relative to the surrounding medium as: $\rho' = \rho - \rho_\infty$ $\frac{\partial \rho}{\partial t} = \frac{\partial \rho'}{\partial t}$
- Then, we subtract $\partial^2(\rho' c_\infty^2)/\partial x_i^2$ from each side of the above equation to yield:
 - c_∞ – ambient speed of sound outside the flow (constant)

$$\Rightarrow \frac{\partial^2(\rho_\infty + \rho')}{\partial t^2} - \frac{\partial^2(\rho' c_\infty^2)}{\partial x_i^2} = \frac{\partial^2(\rho v_i v_j + p_{ij})}{\partial x_i \partial x_j} - \frac{\partial^2(\rho' c_\infty^2)}{\partial x_i^2} \Rightarrow \frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \frac{\partial^2(\rho v_i v_j + p_{ij} - c_\infty^2(\rho - \rho_\infty)\delta_{ij})}{\partial x_i \partial x_j}$$

Lighthill's wave equation

$$\frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}$$

Quadrupole...

Lighthill's stress tensor

$$T_{ij} = \rho v_i v_j + [(p - p_\infty) - (\rho - \rho_\infty)c_\infty^2]\delta_{ij} - \sigma_{ij}$$

Lighthill's acoustic analogy

$$\underbrace{\frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2}}_{\text{Acoustic waves propagation}} = \underbrace{\frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}}_{\text{"Source" term}}$$

Acoustic waves propagation
in a uniform medium with speed c_∞
 ρ' is the dependent variable

"Source" term
volume effects that
generate acoustic waves

$$T_{ij} = \underbrace{\rho v_i v_j}_{\text{Reynolds stresses}} + \underbrace{[(p - p_\infty) - (\rho - \rho_\infty)c_\infty^2] \delta_{ij}}_{\text{Non-homentropic effects}} - \underbrace{\sigma_{ij}}_{\text{Viscosity effects}}$$

- ❑ Lighthill's equation represents **an analogy** for the waves that radiate from a **finite volume of turbulent flow (no solid surfaces)** into a surrounding stationary medium
- ❑ **Acoustic wave propagation is defined with c_∞** – speed of sound in the **stationary medium**
 - NOT in the turbulent flow region, where it could have a very different value due to compressible flow effects, combustion, etc.
 - This choice is *appropriate* because the **objective is to evaluate the acoustic waves that propagate away from the region of turbulent flow in the stationary medium**
- ❑ **Source term** defines the *coupling of turbulence*, or fluid perturbations in the turbulent region, with the sound waves in the far field

Lighthill's acoustic analogy

$$\underbrace{\frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2}}_{\text{Acoustic waves propagation}} = \underbrace{\frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}}_{\text{"Source" term}}$$

Acoustic waves propagation
in a uniform medium with speed c_∞
 ρ' is the dependent variable

"Source" term
volume effects that
generate acoustic waves

$$T_{ij} = \underbrace{\rho v_i v_j}_{\substack{\text{Reynolds stresses} \\ \text{Non-linear effects}}} + \underbrace{[(p - p_\infty) - (\rho - \rho_\infty)c_\infty^2] \delta_{ij}}_{\substack{\text{Non-homentropic} \\ \text{effects}}} - \underbrace{\sigma_{ij}}_{\substack{\text{Viscosity} \\ \text{effects}}}$$

❑ Recall, **NO APPROXIMATIONS** have been made!

❑ In **linear acoustics**:

- $\rho' = \rho - \rho_\infty = (p - p_\infty)/c_\infty^2$
- Nonlinear interaction ignored, $\rho v_i v_j = 0$
- Viscous effects ignored, $\sigma_{ij} = 0$

$$\left. \begin{array}{l} \text{Nonlinear interaction ignored, } \rho v_i v_j = 0 \\ \text{Viscous effects ignored, } \sigma_{ij} = 0 \end{array} \right\} \Rightarrow T_{ij} = 0$$

❑ **Lighthill's equation** includes the effect of turbulence that **cannot be addressed by linear acoustics**

❑ **CAUTION!** Lighthill's equation is *all inclusive*, so it *hides* important details inside the flow region

- Can only **provide insight into sound waves outside the flow**, not in the region of turbulent motion

Limitations of the acoustic analogy

Nearly incompressible flow

$$\frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}$$

$$T_{ij} = \rho v_i v_j + [(p - p_\infty) - (\rho - \rho_\infty) c_\infty^2] \delta_{ij} - \sigma_{ij}$$

- ❑ Commonly, T_{ij} cannot be determined, and so *approximations* are needed to estimate terms that couple effectively with the sound field
 - **CAUTION!** difficulty may arise from the presence of *large fluctuations* of the source terms that are *weakly coupled* to the sound waves and so *effectively do not radiate any sound*
- ❑ Consider an **almost incompressible flow** ($\partial v_i / \partial x_i \approx 0$) with **velocity fluctuations dominated by turbulence** at **high-Reynolds** (i.e., viscous terms are *negligible*, $\sigma_{ij} \approx 0$)
 - Assume homentropic flow, $(p - p_\infty) = (\rho - \rho_\infty) c_\infty^2$, as well as $\rho' \ll \rho_\infty = \text{const.}$, and $v_i^2 \ll c_\infty^2$

$$\frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j} \Rightarrow \frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \rho_\infty \frac{\partial^2 (v_i v_j)}{\partial x_i \partial x_j} \longrightarrow \frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \rho_\infty \frac{\partial v_j}{\partial x_i} \frac{\partial v_i}{\partial x_j}$$

For an almost incompressible flow, Lighthill's source term is **nonlinear** and determined by the **velocity gradients**

$$\frac{\partial^2 (v_i v_j)}{\partial x_i \partial x_j} = \frac{\partial}{\partial x_i} \left(v_i \frac{\partial v_j}{\partial x_j} + v_j \frac{\partial v_i}{\partial x_j} \right) = \frac{\partial}{\partial x_i} \left(v_i \frac{\partial v_j}{\partial x_j} \right) + \frac{\partial v_j}{\partial x_i} \frac{\partial v_i}{\partial x_j} + v_j \frac{\partial^2 v_i}{\partial x_i \partial x_j} \approx \frac{\partial v_j}{\partial x_i} \frac{\partial v_i}{\partial x_j}$$

$\partial v_i / \partial x_i \approx 0$

Limitations of the acoustic analogy

Nearly incompressible flow

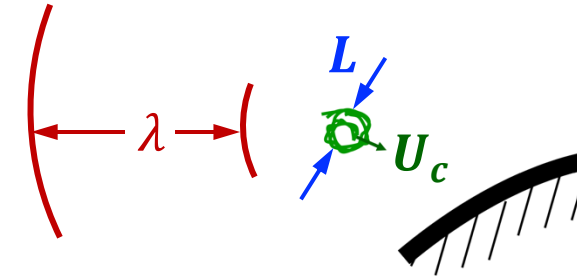
$$\frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}$$

$$T_{ij} = \rho v_i v_j + [(p - p_\infty) - (\rho - \rho_\infty) c_\infty^2] \delta_{ij} - \sigma_{ij}$$

$$\frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \rho_\infty \frac{\partial v_j}{\partial x_i} \frac{\partial v_i}{\partial x_j}$$

- Recall we found that at low Mach flows ($M \ll 1$) the **length scale of the turbulent eddies** in the flow is **much less** than the **acoustic wavelength** at the same frequency
 - U_c - average convection velocity; typically for BLs, $U_c = 0.6 - 0.8 U_\infty$

$$\text{Length scale of turbulent eddies} \quad \frac{U_c}{f} = L \ll \lambda = \frac{c_0}{f} \quad \text{Acoustic wavelength}$$



- Since the source is *nonlinear*, it will generate perturbations at the sum and difference of two interacting velocity perturbations
- Thus, the **radiated sound field will occur at a different frequency** from that of the velocity perturbation observed in the flow!

Flow noise is generated by *nonlinear* interaction

Linear approximations **NOT appropriate!**

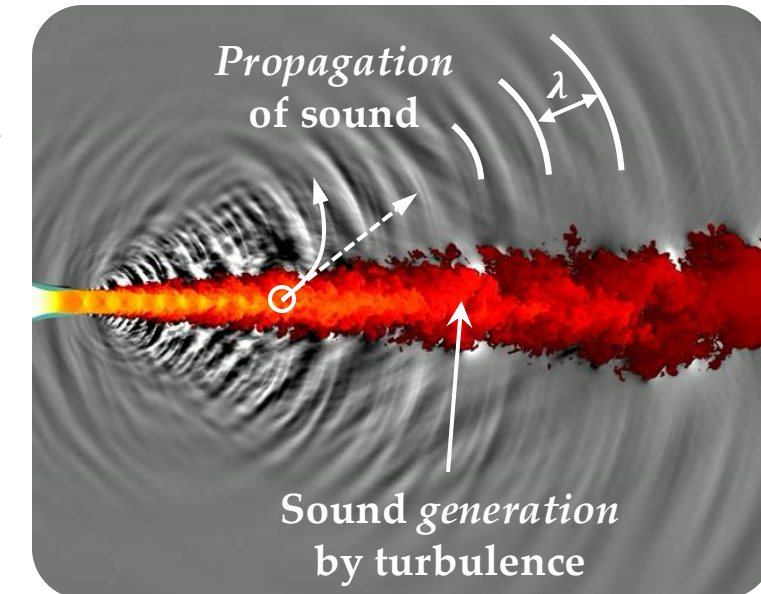
Limitations of the acoustic analogy

Uniform flow

$$\frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}$$

$$T_{ij} = \rho v_i v_j + [(p - p_\infty) - (\rho - \rho_\infty) c_\infty^2] \delta_{ij} - \sigma_{ij}$$

- ❑ In aeroacoustic applications, we cannot *always* assume $v_i^2 \ll c_\infty^2$
 - Local sound speed may vary *significantly* from that outside the flow
- ❑ **Two effects are present in turbulent flow:**
 - **Sound generation** by turbulence in a turbulent eddy
 - **Propagation of sound** from the source element, through the *nonuniform* flow, to acoustic far field
 - Causes acoustic waves to **bend** as they propagate (hidden in T_{ij})
- ❑ Although Lighthill's tensor includes these two effects, **separating sound propagation** effects from **sound generation** in T_{ij} is **not trivial**...
- ❑ We shall illustrate this subtle in the next slides



Limitations of the acoustic analogy

Uniform flow

$$\frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}$$

$$T_{ij} = \rho v_i v_j + [(p - p_\infty) - (\rho - \rho_\infty) c_\infty^2] \delta_{ij} - \sigma_{ij}$$

- ❑ Consider **turbulence in a uniform moving medium**, where: $v_i = U_i^{(\infty)} + w_i$
 - $U_i^{(\infty)}$ – constant uniform velocity (everywhere)
 - w_i – fluctuating velocity
- ❑ Lighthill's stress tensor can therefore be written as: $T_{ij} = \rho (U_i^{(\infty)} + w_i) (U_j^{(\infty)} + w_j) + p_{ij} - \rho' c_\infty^2 \delta_{ij}$
 - Expanding the product of velocities term by term yields:

$$T_{ij} = \rho (U_i^{(\infty)} U_j^{(\infty)} + U_j^{(\infty)} w_i + U_i^{(\infty)} w_j + w_i w_j) + p_{ij} - \rho' c_\infty^2 \delta_{ij}$$

- ❑ Therefore, Lighthill's source term becomes:

$$\frac{\partial^2 T_{ij}}{\partial x_i \partial x_j} = U_i^{(\infty)} U_j^{(\infty)} \frac{\partial^2 \rho}{\partial x_i \partial x_j} + 2 U_j^{(\infty)} \underbrace{\frac{\partial^2 (\rho w_i)}{\partial x_i \partial x_j}} + \frac{\partial^2}{\partial x_i \partial x_j} (\rho w_i w_j + p_{ij} - \rho' c_\infty^2 \delta_{ij})$$

$$\text{Continuity eq.: } \frac{\partial \rho}{\partial t} + \frac{\partial (\rho v_i)}{\partial x_i} = 0 \Rightarrow \frac{\partial (\rho w_i)}{\partial x_i} = -\frac{\partial \rho}{\partial t} - U_i^{(\infty)} \frac{\partial \rho}{\partial x_i}$$

$$\Rightarrow \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j} = U_i^{(\infty)} U_j^{(\infty)} \frac{\partial^2 \rho}{\partial x_i \partial x_j} - 2 U_j^{(\infty)} \frac{\partial}{\partial x_j} \left(\frac{\partial \rho}{\partial t} + U_i^{(\infty)} \frac{\partial \rho}{\partial x_i} \right) + \frac{\partial^2}{\partial x_i \partial x_j} (\rho w_i w_j + p_{ij} - \rho' c_\infty^2 \delta_{ij})$$

Limitations of the acoustic analogy

Uniform flow

$$\frac{\partial^2 \rho'}{\partial t^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}$$

$$T_{ij} = \rho v_i v_j + [(p - p_\infty) - (\rho - \rho_\infty) c_\infty^2] \delta_{ij} - \sigma_{ij}$$

□ Re-arranging, we obtain:

$$\frac{\partial^2 T_{ij}}{\partial x_i \partial x_j} = -U_i^{(\infty)} U_j^{(\infty)} \frac{\partial^2 \rho}{\partial x_i \partial x_j} - 2U_j^{(\infty)} \frac{\partial^2 \rho}{\partial t \partial x_j} + \frac{\partial^2}{\partial x_i \partial x_j} (\rho w_i w_j + p_{ij} - \rho' c_\infty^2 \delta_{ij})$$

$\rho' = \rho - \rho_\infty$
 $p_{ij} = (p - p_\infty) \delta_{ij} - \sigma_{ij}$

□ Since $\rho = \rho_\infty + \rho'$, where $\rho_\infty = \text{const.}$, Lighthill's equation reads:

$$\Rightarrow \underbrace{\frac{\partial^2 \rho'}{\partial t^2} + 2U_j^{(\infty)} \frac{\partial^2 \rho}{\partial t \partial x_j} + U_i^{(\infty)} U_j^{(\infty)} \frac{\partial^2 \rho}{\partial x_i \partial x_j}}_{\text{Material derivative}} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \frac{\partial^2}{\partial x_i \partial x_j} (\rho w_i w_j + p_{ij} - \rho' c_\infty^2 \delta_{ij})$$

Material derivative $\frac{D_\infty \rho'}{Dt} = \frac{\partial \rho'}{\partial t} + U_j^{(\infty)} \frac{\partial \rho'}{\partial x_j}$

Source term depends on velocity perturbations

Mean flow causes propagation effects!

Wave equation

for a stationary fluid from a frame of reference moving with the freestream

$$\frac{D_\infty^2 \rho'}{Dt^2} - c_\infty^2 \frac{\partial^2 \rho'}{\partial x_i^2} = \frac{\partial^2}{\partial x_i \partial x_j} (\rho w_i w_j + p_{ij} - \rho' c_\infty^2 \delta_{ij})$$

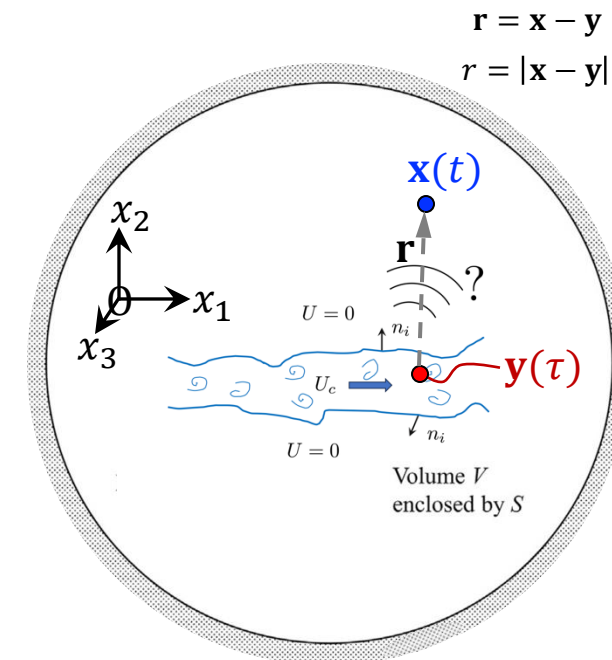
Lighthill's equation can be applied in a uniform flow where the acoustic waves satisfy a **convective wave equation**

Solution to Lighthill's analogy

- ❑ A solution to Lighthill's analogy can be obtained by using the method of Green's functions
- ❑ Consider a **stationary uniform medium** (with volume V that is enclosed by the surface S), which includes a **bounded region of unsteady flow**
 - Linear acoustics won't apply in the flow region
- ❑ We start by writing Lighthill's analogy in terms of $\mathbf{y}$ and τ as follows:
 - Same structure as the inhomogeneous wave equation, but with $p' \Rightarrow \rho' c_\infty^2$

$$\boxed{\frac{1}{c_\infty^2} \frac{\partial^2(\rho' c_\infty^2)}{\partial \tau^2} - \frac{\partial^2(\rho' c_\infty^2)}{\partial y_i^2} = \frac{\partial^2 T_{ij}}{\partial y_i \partial y_j}}$$

$$T_{ij} = \rho v_i v_j + [(p - p_\infty) - (\rho - \rho_\infty) c_\infty^2] \delta_{ij} - \sigma_{ij}$$



Solution to Lighthill's analogy

□ As before...

➤ Take Green's function equation:

$$\frac{1}{c_\infty^2} \frac{\partial^2 G}{\partial \tau^2} - \frac{\partial^2 G}{\partial y_i^2} = \delta(\mathbf{x} - \mathbf{y}) \delta(t - \tau) \quad \times \quad \rho'(\mathbf{y}, \tau) c_\infty^2$$

➤ Take Lighthill's equation:

$$\frac{1}{c_\infty^2} \frac{\partial^2 (\rho' c_\infty^2)}{\partial \tau^2} - \frac{\partial^2 (\rho' c_\infty^2)}{\partial y_i^2} = \frac{\partial^2 T_{ij}}{\partial y_i \partial y_j} \quad \times \quad G(\mathbf{x}, t | \mathbf{y}, \tau)$$

} (-) Subtract

➡

$$\frac{1}{c_\infty^2} \left(\rho' c_\infty^2 \frac{\partial^2 G}{\partial \tau^2} - G \frac{\partial^2 (\rho' c_\infty^2)}{\partial \tau^2} \right) - \left(\rho' c_\infty^2 \frac{\partial^2 G}{\partial y_i^2} - G \frac{\partial^2 (\rho' c_\infty^2)}{\partial y_i^2} \right) = \delta(\mathbf{x} - \mathbf{y}) \delta(t - \tau) \rho'(\mathbf{y}, \tau) c_\infty^2 - G \frac{\partial^2 T_{ij}}{\partial y_i \partial y_j}$$

□ Integrating over volume V and time τ (as we previously did) results in:

$$\begin{aligned} \rho'(\mathbf{x}, t) c_\infty^2 = & \int_{-T}^T \int_S \left(\rho'(\mathbf{y}, \tau) c_\infty^2 \frac{\partial G}{\partial y_i} - G \frac{\partial (\rho'(\mathbf{y}, \tau) c_\infty^2)}{\partial y_i} \right) n_i dS(\mathbf{y}) d\tau \\ & + \int_{-T}^T \int_V G \left(\frac{\partial^2 T_{ij}(\mathbf{y}, \tau)}{\partial y_i \partial y_j} \right) dV(\mathbf{y}) d\tau \end{aligned}$$

- Providing $\mathbf{x}$ is inside V and $-T < t < T$

Solution to Lighthill's analogy

$$\rho'(\mathbf{x}, t)c_\infty^2 = \int_{-T}^T \int_S \left(\rho'(\mathbf{y}, \tau)c_\infty^2 \frac{\partial G}{\partial y_i} - G \frac{\partial(\rho'(\mathbf{y}, \tau)c_\infty^2)}{\partial y_i} \right) n_i dS(\mathbf{y}) d\tau + \int_{-T}^T \int_V G \left(\frac{\partial^2 T_{ij}(\mathbf{y}, \tau)}{\partial y_i \partial y_j} \right) dV(\mathbf{y}) d\tau$$

- ❑ The equation above, while correct, is a **dangerous result...**
- ❑ If the **turbulent region** (the source region where $T_{ij} \neq 0$) is **very small**, we can *neglect* the propagation times between different points $\mathbf{y}$ inside the volume and the observer at $\mathbf{x}$
 - In such a scenario, Green's function G can then be taken outside the volume integral, leading to a **surface integral of the source term** (following the divergence theorem): $\int_{-T}^T \int_S (\partial T_{ij}(\mathbf{y}, \tau) / \partial y_j) n_j dS(\mathbf{y}) d\tau$
- ❑ **BUT! In the absence of a surface in the flow**, the integrand of the source term $(\partial T_{ij}(\mathbf{y}, \tau) / \partial y_j)$ comes from the surface integral placed **at the end of the turbulent flow region – where $T_{ij} = 0$**
 - That's a trivial solution of no physical meaning... in which the equation above gives $0 = 0$
- ❑ Therefore, to get the correct result for an observer at $\mathbf{x}$, one must:
 - Evaluate the integrand *precisely* at the correct source times τ
 - Carry out the integration with *no rounding errors*

$$\int_{-T}^T \int_V G \left(\frac{\partial^2 T_{ij}(\mathbf{y}, \tau)}{\partial y_i \partial y_j} \right) dV(\mathbf{y}) d\tau$$

Challenging! numerical integration errors can exceed the contribution from the source terms...

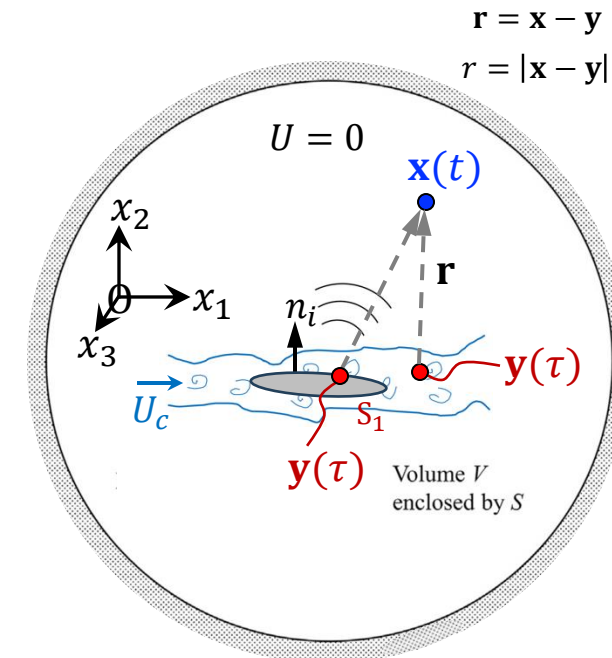
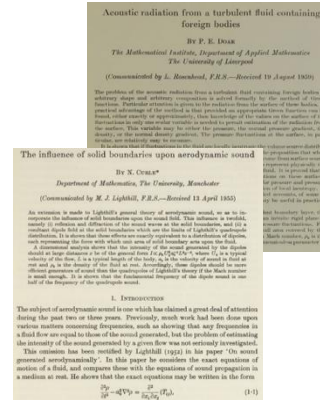
Curle's theorem

- ❑ Lighthill recognized the issue with integrating and proposed an important simplification, identifying the leading order source terms in the volume integral, so that it could be evaluated without requiring exact retarded time calculations
- ❑ This concept was later extended by both [Curle](#) (1955) and [Doak](#) (1960) to apply to turbulent flow in the presence of surfaces
- ❑ Consider a **stationary uniform medium** that includes a **bounded region of turbulent flow in the presence of surfaces**
- ❑ We will derive Curle's equation by using the expansion (Doak's approach):

$$\frac{\partial}{\partial \mathbf{y}_i} \left(G \frac{\partial T_{ij}}{\partial \mathbf{y}_j} \right) - \frac{\partial}{\partial \mathbf{y}_j} \left(T_{ij} \frac{\partial G}{\partial \mathbf{y}_i} \right) = G \frac{\partial^2 T_{ij}}{\partial \mathbf{y}_i \partial \mathbf{y}_j} - T_{ij} \frac{\partial^2 G}{\partial \mathbf{y}_i \partial \mathbf{y}_j}$$

Substitute into
volume integral

$$\int_{-T}^T \int_V G \left(\frac{\partial^2 T_{ij}(\mathbf{y}, \tau)}{\partial \mathbf{y}_i \partial \mathbf{y}_j} \right) dV(\mathbf{y}) d\tau$$



Curle, N. (1955) "The influence of solid boundaries upon aerodynamic sound", *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 231(1187), 505–514.

Doak, P. E. (1960) "Acoustic radiation from a turbulent fluid containing foreign bodies", *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 254(1276), 129–146.

Curle's theorem

$$\frac{\partial}{\partial \mathbf{y}_i} \left(G \frac{\partial T_{ij}}{\partial \mathbf{y}_j} \right) - \frac{\partial}{\partial \mathbf{y}_j} \left(T_{ij} \frac{\partial G}{\partial \mathbf{y}_i} \right) = G \frac{\partial^2 T_{ij}}{\partial \mathbf{y}_i \partial \mathbf{y}_j} - T_{ij} \frac{\partial^2 G}{\partial \mathbf{y}_i \partial \mathbf{y}_j}$$

$$\int_{-T}^T \int_V G \left(\frac{\partial^2 T_{ij}(\mathbf{y}, \tau)}{\partial \mathbf{y}_i \partial \mathbf{y}_j} \right) dV(\mathbf{y}) d\tau = \int_{-T}^T \int_V \left(\frac{\partial^2 G}{\partial \mathbf{y}_i \partial \mathbf{y}_j} \right) T_{ij}(\mathbf{y}, \tau) dV(\mathbf{y}) d\tau + \int_{-T}^T \int_V \frac{\partial}{\partial \mathbf{y}_i} \left(G \frac{\partial T_{ij}(\mathbf{y}, \tau)}{\partial \mathbf{y}_j} \right) dV(\mathbf{y}) d\tau - \int_{-T}^T \int_V \frac{\partial}{\partial \mathbf{y}_j} \left(T_{ij}(\mathbf{y}, \tau) \frac{\partial G}{\partial \mathbf{y}_i} \right) dV(\mathbf{y}) d\tau$$

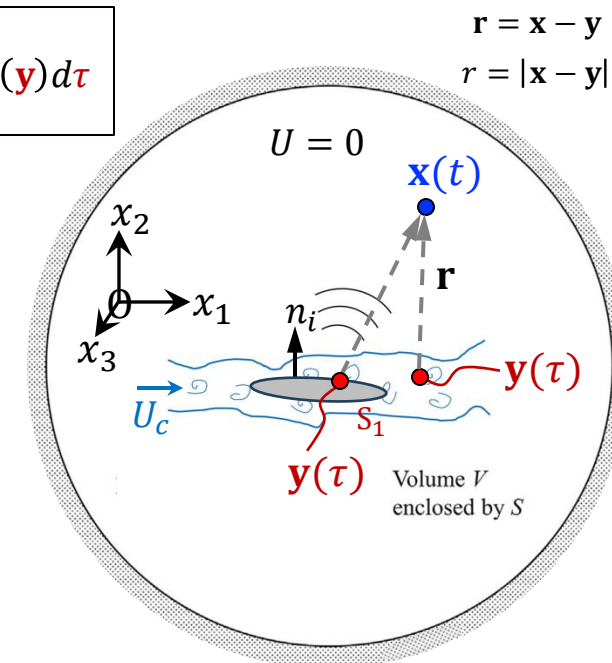
□ Using the *divergence theorem* on the last two terms, and note that $T_{ij} = T_{ji}$, we obtain:

➤ n_i – unit vector normal to the surface of $\mathbf{y}$ pointing *into* the volume

$$\Rightarrow \int_{-T}^T \int_V G \left(\frac{\partial^2 T_{ij}(\mathbf{y}, \tau)}{\partial \mathbf{y}_i \partial \mathbf{y}_j} \right) dV(\mathbf{y}) d\tau = \int_{-T}^T \int_V \left(\frac{\partial^2 G}{\partial \mathbf{y}_i \partial \mathbf{y}_j} \right) T_{ij}(\mathbf{y}, \tau) dV(\mathbf{y}) d\tau + \int_{-T}^T \int_S \left(T_{ij}(\mathbf{y}, \tau) \frac{\partial G}{\partial \mathbf{y}_i} - G \frac{\partial T_{ij}(\mathbf{y}, \tau)}{\partial \mathbf{y}_i} \right) n_j dS(\mathbf{y}) d\tau$$

□ Substituting into the [solution to Lighthill's equation](#) yields:

$$\begin{aligned} \rho'(\mathbf{x}, t) c_\infty^2 = & \int_{-T}^T \int_S \left((\rho' c_\infty^2 \delta_{ij} + T_{ij}) \frac{\partial G}{\partial \mathbf{y}_i} - G \frac{\partial (\rho' c_\infty^2 \delta_{ij} + T_{ij})}{\partial \mathbf{y}_i} \right) n_j dS(\mathbf{y}) d\tau \\ & + \int_{-T}^T \int_V \left(\frac{\partial^2 G}{\partial \mathbf{y}_i \partial \mathbf{y}_j} \right) T_{ij}(\mathbf{y}, \tau) dV(\mathbf{y}) d\tau \end{aligned}$$



$$T_{ij} = \rho v_i v_j + [(p - p_\infty) - (\rho - \rho_\infty) c_\infty^2] \delta_{ij} - \sigma_{ij}$$

$$\frac{\partial(\rho v_i)}{\partial t} + \frac{\partial(\rho v_i v_j + p_{ij})}{\partial x_j} = 0$$

Curle's theorem

- Using the definition of *Lighthill's stress tensor*, we note that: $\rho' c_\infty^2 \delta_{ij} + T_{ij} = p_{ij} + \rho v_i v_j$
- Moreover, using the momentum equation, we obtain the following:

$$\rho'(\mathbf{x}, t) c_\infty^2 = \int_{-T}^T \int_S \left((p_{ij} + \rho v_i v_j) \frac{\partial G}{\partial y_i} + G \frac{\partial(\rho v_j)}{\partial \tau} \right) n_j dS(\mathbf{y}) d\tau + \int_{-T}^T \int_V \left(\frac{\partial^2 G}{\partial y_i \partial y_j} \right) T_{ij}(\mathbf{y}, \tau) dV(\mathbf{y}) d\tau$$

(II) Force per unit area $p_{ij} n_j$ applied to the fluid by surface

(III) Momentum flux across surface

(I) Rate of change of mass flux through the surface

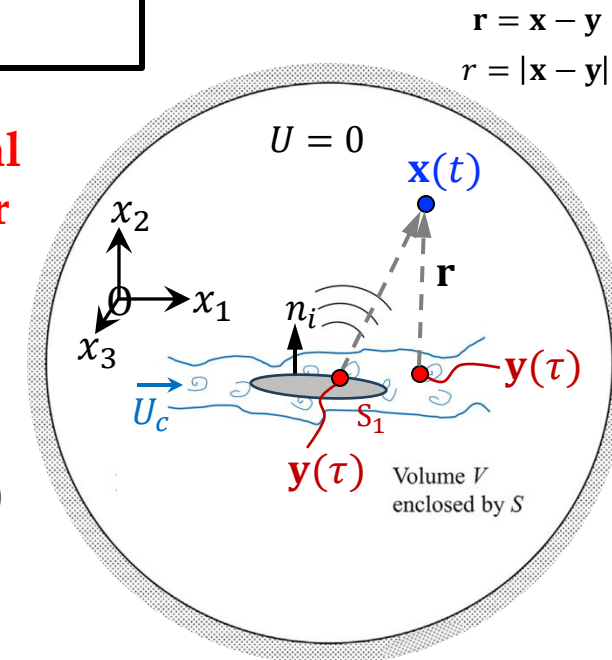
(IV) Volume integral of Lighthill's tensor distribution throughout the fluid

(I) and (IV) are the **additional nonlinear noise terms!!!**

- This is a more general solution than the linear acoustic solution to $\square p' = 0$

Restrictions:

- Stationary surfaces bounding the fluid (can be impermeable)
- Stationary medium outside the turbulent flow region, $U = 0$



Curle's theorem

$$\rho'(\mathbf{x}, t) c_\infty^2 = \int_{-T}^T \int_S \left((p_{ij} + \rho v_i v_j) \frac{\partial G}{\partial y_i} + G \frac{\partial(\rho v_j)}{\partial \tau} \right) n_j dS(\mathbf{y}) d\tau + \int_{-T}^T \int_V \left(\frac{\partial^2 G}{\partial y_i \partial y_j} \right) T_{ij}(\mathbf{y}, \tau) dV(\mathbf{y}) d\tau$$

- Result depends on a Green's function that needs to be a solution of: $\frac{1}{c_0^2} \frac{\partial^2 G}{\partial \tau^2} - \frac{\partial^2 G}{\partial y_i^2} = \delta(\mathbf{x} - \mathbf{y}) \delta(t - \tau)$

and satisfy a **causality condition**: $G = \frac{\partial G}{\partial \tau} = 0$ for $t < \tau$

- Recall the **free-field Green's function**: $G_o(\mathbf{x}, t | \mathbf{y}, \tau) = \frac{\delta(t - |\mathbf{x} - \mathbf{y}|/c_o - \tau)}{4\pi|\mathbf{x} - \mathbf{y}|}$

$$\text{Symmetry property} \begin{cases} \frac{\partial G_o}{\partial x_i} = -\frac{\partial G_o}{\partial y_i} \\ \frac{\partial^2 G_o}{\partial y_i \partial y_j} = \frac{\partial^2 G_o}{\partial x_i \partial x_j} \end{cases}$$

- We can therefore rewrite the solution in terms of G_o :

$$\rho'(\mathbf{x}, t) c_\infty^2 = \int_{-T}^T \int_S \frac{\partial(\rho v_j)}{\partial \tau} G_o n_j dS(\mathbf{y}) d\tau - \frac{\partial}{\partial x_i} \int_{-T}^T \int_S (p_{ij} + \rho v_i v_j) G_o n_j dS(\mathbf{y}) d\tau + \frac{\partial^2}{\partial x_i \partial x_j} \int_{-T}^T \int_V T_{ij}(\mathbf{y}, \tau) G_o dV(\mathbf{y}) d\tau$$

Curle's theorem

- Substitute the expression of G_o and perform the time integrals yields:

$$\rho'(\mathbf{x}, t) c_\infty^2 = \int_S \left[\frac{\partial(\rho v_j)}{\partial \tau} \right]_{\tau=\tau^*} \frac{n_j dS(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|} - \frac{\partial}{\partial x_i} \int_S [p_{ij} + \rho v_i v_j]_{\tau=\tau^*} \frac{n_j dS(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|} + \frac{\partial^2}{\partial x_i \partial x_j} \int_V T_{ij}(\mathbf{y}, \tau^*) \frac{dV(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|}$$

Retarded time

$$\tau^* = t - \frac{|\mathbf{x} - \mathbf{y}|}{c_0}$$

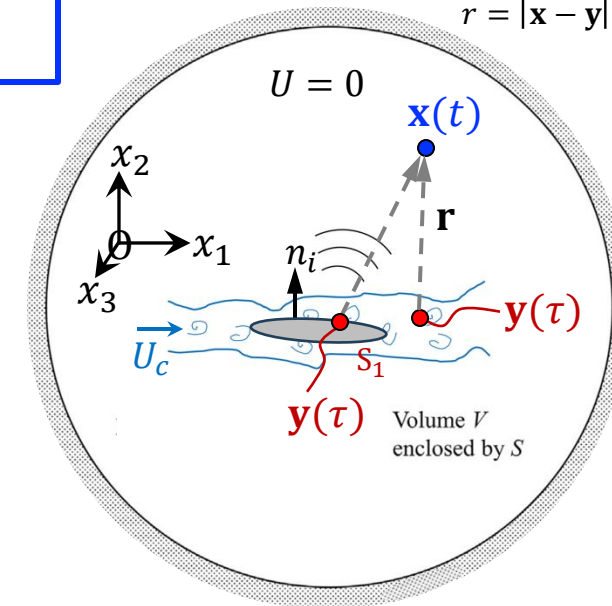
$$\mathbf{r} = \mathbf{x} - \mathbf{y}$$

$$r = |\mathbf{x} - \mathbf{y}|$$

- Note that source terms are evaluated at the **correct retarded times**
- Surfaces are stationary so $\mathbf{y}$ is independent of τ , and fluctuations occurring at different parts of the flow at the same source time τ will contribute to received signal at $\mathbf{x}$ at different reception times t

Retarded time is very important to accurately predict far-field sound from flow noise sources!

- For $V \rightarrow 0$, retarded time differences can be ignored, thus the volume integral can be evaluated at a fixed source time



$$\int_{-\infty}^{\infty} f(\tau) \delta(t - r/c_0 - \tau) d\tau = f(\tau^*)$$

Monopole, dipole, and quadrupole

□ We notice the following breakdown:

$$\begin{aligned}
 \rho'(\mathbf{x}, t)c_\infty^2 = & \overbrace{\int_S \left[\frac{\partial(\rho v_j)}{\partial \tau} \right]_{\tau=\tau^*} \frac{n_j dS(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|}}^{(\rho'c_\infty^2)_{\text{monopole}}} - \overbrace{\frac{\partial}{\partial x_i} \int_S [p_{ij} + \rho v_i v_j]_{\tau=\tau^*} \frac{n_j dS(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|}}^{(\rho'c_\infty^2)_{\text{dipole}}} \\
 & + \underbrace{\frac{\partial^2}{\partial x_i \partial x_j} \int_V T_{ij}(\mathbf{y}, \tau^*) \frac{dV(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|}}_{(\rho'c_\infty^2)_{\text{quadrupole}}}
 \end{aligned}$$

Monopole, dipole, and quadrupole

□ Monopole

- Surface on which the fluid moves with a normal velocity $u_n = v_i n_i$
- Due to **unsteady mass flux** across surface
 - Notice that the mass flux is *zero* for rigid or impermeable surfaces
- **Omnidirectional** in acoustic far-field $|\mathbf{x}| \gg |\mathbf{y}|$
- Driven by net rate of mass flux through the surface – **independent of shape!**
 - Thus, if acoustically compact, we get the same acoustic field in the far field for the same mass flux injection



$$(\rho'(\mathbf{x}, t) c_\infty^2)_{\text{monopole}} = \int_S \left[\frac{\partial(\rho u_n)}{\partial \tau} \right]_{\tau=\tau^*} \frac{dS(\mathbf{y})}{4\pi|\mathbf{x} - \mathbf{y}|}$$

- If acoustically compact (surface is sufficiently small) in the acoustic far-field we obtain:

$$(\rho'(\mathbf{x}, t) c_\infty^2)_{\text{monopole}} = \frac{1}{4\pi\mathbf{x}} \left[\frac{\partial \dot{M}}{\partial \tau} \right]_{\tau=\tau^*} \quad \text{where} \quad \dot{M} = \int_S \rho u_n dS(\mathbf{y}) \quad \text{Mass flow rate out of surface}$$